01270004 — Quantum Scattering Theory

Lecture Notes with Project Results (Tasks 1–6)

Version 1.0

Based on Lectures 1–11 (28 Jul – 1 Sep 2026) and the project tasks

Coverage: Lectures 1–11 and project parts 1–6
Topics: 1D quantum mechanics recap, Dirac delta, free and bound states, short-ranged potentials, finite differences, transmission/reflection, wavepackets, formal scattering theory ($\hat{S}$, $\hat{T}$, Lippmann–Schwinger), variational methods, non-Hermitian QM, complex scaling

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Contents

Legend: Box-Type Reference **2**

1 Overview and Roadmap **3**

2 Lecture 1 — Classical and Quantum Description in 1D **4**
2.1 Classical particle . . . 4
2.2 Quantum description . . . 4

3 Lecture 2 — Free Particle, Eigenfunctions, and the Dirac Delta **4**
3.1 Free propagation and Gaussian wavepackets . . . 4
3.2 Eigenfunctions in the continuum: not square integrable, degenerate . . . 5
3.3 Bound spectrum: expansion in eigenstates . . . 5
3.4 Dirac delta as the continuous analogue of $\delta_{jj'}$ . . . 5
3.5 Project Task 1 — Free Gaussian Wavepackets . . . 6
3.6 Project Task 2 — Dirac Delta from the Dirichlet Kernel . . . 8

4 Lecture 3 — Short-Ranged Potentials and Scattering Boundary Conditions **9**
4.1 Continuum solutions and scattering boundary conditions . . . 10
4.2 Method: finite differences and the Neuhauser trick . . . 10
4.3 Project Task 3 — Transmission Profile . . . 10

5 Lectures 4–5 — Spectrum, Wavepackets, Asymptotics **13**
5.1 Project Task 4 — Wavepacket Scattering (TDSE, Finite Differences) . . . 13

6 Lectures 5–6 — Formal Scattering Theory in the Time Domain **17**
6.1 Construction of $\hat{S}$ . . . 18
6.2 The iε prescription and the Lippmann–Schwinger equation . . . 18

7 Lecture 7 — Green Functions, Born Series, and $T(p)$, $R(p)$ from the LSE **19**

8 Lecture 8 — The $\hat{S}$ Matrix, the $\hat{T}$ Operator, Energy Conservation **20**

9 Lecture 9 — Transmission and Reflection Probabilities; Variational Bound States **20**
9.1 From $\hat{S}$ to T and R . . . 20
9.2 Project part 5 — Variational (basis-set) method for bound states . . . 21
9.3 Project Task 5 — Variational Bound States . . . 21

10 Lecture 10 — Non-Hermitian Quantum Mechanics and Siegert States **23**

11 Lecture 11 — Complex Scaling, Resonances and the Breit–Wigner Profile **24**
11.1 Why we need something computable . . . 24
11.2 Project Task 6 — Complex Scaling (NHQM) . . . 25

12 Consolidated Summary and Self-Check **28**

Legend: Box-Type Reference

All mathematical content in these notes is displayed in colour-coded boxes. The table below gives the full reference; subsequent new content must follow this convention.

Environment	Colour	Usage
`defbox`	Blue	Definitions; formal notation
`thmbox`	Crimson	Theorems; Lemmas
`proofbox`	Gray	Full or sketched proofs
`corbox`	Teal	Corollaries
`propbox`	Violet	Propositions
`tipbox`	Amber	Remarks; Notes; Recalls
`warnbox`	Red	Warnings; important notices; exam scope
`flowbox`	Dark orange	Key formulae; summary boxes; recurrences
`exbox`	Warm umber	Worked examples
`methodbox`	Steel blue	Proof strategies; algorithms; methods

Remark — Primary Reference

M. Šindelka, *An Introduction to Scattering Theory*, lecture notes, arXiv:2204.03651 (2022), used only as a cross-check; every result below appears in Lectures 1–11. These notes follow the lecture structure and are not tied to any particular chapter ordering of the textbook.

1 Overview and Roadmap

Lecture 1 fixed the plan of the course: *basic scattering theory*, i.e. the quantum theory of collisions (particle physics, chemical reactions, light–matter interaction, scattering of light in waveguides). The pace is slow (understanding over quantity); assessment is a **project plus an oral interview**. The material splits into three parts, following the handwritten note pages:

1. recap of mathematical tools and Hermitian quantum mechanics (L1–L3);
2. 1D scattering, practically (numerics, wavepackets) and theoretically (formal scattering theory) (L3–L9);
3. deeper understanding of resonances with non-Hermitian quantum mechanics (NHQM) (L10–L11).

Key Formula Summary — Conventions Used Throughout

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) = -\frac{\hbar^2}{2m}\partial_{xx} + V(x), \qquad \hat{p} = -\mathrm{i}\hbar\,\partial_x, \qquad \mathrm{i}\hbar\,\partial_t\Psi(t,x) = \hat{H}\Psi(t,x).$$

In all numerical work $\hbar = m = 1$. Free continuum states $\psi_p^{\mathrm{F}}(x) = e^{\mathrm{i}px/\hbar}/\sqrt{2\pi\hbar}$, $E_p = p^2/2m$. $\pm\mathrm{i}\varepsilon$ with $\varepsilon \to 0^+$ always means the limit taken *after* all integrals.

Remark — Project map

The six computational tasks are woven through the lectures; the *results* of each task are given at the end of the corresponding lecture section as "Project Task n:

Task	Content	Lecture
1	free Gaussian wavepackets, animation	L2
2	regularised Dirac δ (Dirichlet kernel)	L2–L3
3	transmission/reflection, resonance peaks (finite differences)	L3–L4
4	wavepacket scattering: blocked / partially blocked / pass	L4–L6
5	variational bound states (box basis)	L8–L9
6	complex scaling, Breit–Wigner, resonance wavefunction	L10–L11

Key Formula Summary — Project Setup and Reproducibility

All calculations use $\hbar = m = 1$ and the project potential $V(x) = (0.5x^2 - 0.8)\,e^{-0.1x^2}$ (Tasks 3–6). Code lives in `03_code/` (Python: NumPy, SciPy, Matplotlib); Project results are inserted at the end of the lecture they belong to. Run the scripts in the order `task1`, `task2`, `task3`, `task5`, `task6`, `task6b`, `task4`. Figures go to `04_results/figures`, GIFs to `04_results/animations`, raw numbers to `04_results/data`.

Remark — Interpretation of the project task statements

The task texts for Tasks 3–6 are only given as pictures of the Moodle project pages (results, not instructions), so the tasks were reconstructed from Lectures 3–11: Task 3 = transmission profile and continuum states, Task 4 = solve the TDSE for wavepackets, Task 5 = variational bound states, Task 6 = complex scaling. "FIM" in Task 4 was read as the *finite-difference method* (the method of Lecture 1, Methods #2); the PDE is the TDSE. Every reproduced number below is compared with the values visible on the Moodle pages where such values are shown.

2 Lecture 1 — Classical and Quantum Description in 1D

2.1 Classical particle

A particle on the real line with potential $V(x)$ obeys Newton's law $m\ddot{x} = -V'(x)$, or Hamilton's equations for $H = p^2/2m + V(x)$:

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}, \qquad \dot{p} = -\frac{\partial H}{\partial x} = -V'(x).$$

The initial state is $(x(t_0), p(t_0))$. Example: $V = \frac{1}{2}m\omega^2 x^2$, $m\ddot{x} = -m\omega^2 x$.

2.2 Quantum description

Definition — State, Probability Density, Square Integrability

A pure state is a wavefunction $\Psi : \mathbb{R} \to \mathbb{C}$, equivalently a vector $|\Psi\rangle$ with $\Psi(x) = \langle x|\Psi\rangle$. $|\Psi(x)|^2$ is the probability density, so a physical state is **square integrable**, $\int_{-\infty}^{+\infty} |\Psi|^2\, dx < \infty$, and normalised to $\langle\Psi|\Psi\rangle = 1$. The wavefunction space is $L^2(\mathbb{R})$.

Example — Normalising a Gaussian

$\Psi(x) = e^{-5x^2}$ is legitimate but $\int |\Psi|^2 dx = \sqrt{\pi/10} \neq 1$. Write $\Psi = Ce^{-5x^2}$ with $|C|^2\sqrt{\pi/10} = 1$, i.e. $C = (10/\pi)^{1/4}e^{\mathrm{i}\alpha}$, $\alpha \in \mathbb{R}$. The constant phase $e^{\mathrm{i}\alpha}$ (*global phase factor*) has no physical meaning: it does not change the quantum state. By contrast $\Psi(x) = x^2$ is not square integrable and is not an admissible state.

Dynamics and expectation values. Position and momentum become $\hat{x}\Psi = x\Psi$ and $\hat{p}\Psi = -\mathrm{i}\hbar\Psi'$; time evolution is unitary and follows the TDSE. For any observable $\hat{A}$,

$$\bar{A} = \frac{\langle\Psi|\hat{A}|\Psi\rangle}{\langle\Psi|\Psi\rangle} = \frac{\int dx\, \Psi^*(t,x)\hat{A}\Psi(t,x)}{\int dx\, \Psi^*\Psi}.$$

Remark — Classification of potentials (the three spectral types)

- **Free particle** $V = 0$: energies not quantised, continuum $E \geq 0$.
- **Bound (confining) potentials**, e.g. harmonic or anharmonic oscillator: only quantised levels.
- **Short-ranged potentials**: finitely many bound levels ($E < 0$, possibly none) *and* a continuum $E > 0$.

3 Lecture 2 — Free Particle, Eigenfunctions, and the Dirac Delta

3.1 Free propagation and Gaussian wavepackets

For $V = 0$, with initial state $\varphi(x)$ square integrable, use the Fourier pair

$$\varphi(x) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} dp\, \phi(p)\, e^{\mathrm{i}px/\hbar}, \qquad \phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{+\infty} dx\, \varphi(x)\, e^{-\mathrm{i}px/\hbar}.$$

Because $e^{\mathrm{i}px/\hbar}$ is an eigenfunction of $\hat{H}_0 = \hat{p}^2/2m$ with eigenvalue $p^2/2m$,

$$\Psi(t,x) = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \phi(p)\, e^{\mathrm{i}px/\hbar}\, e^{-\mathrm{i}\frac{p^2}{2m\hbar}t} \;=\; e^{-\mathrm{i}\hat{H}_0 t/\hbar}\varphi. \tag{1}$$

Example — Task 1: free Gaussian wavepacket

Take $\Psi(0,x) = (2\alpha/\pi)^{1/4} e^{-\alpha x^2} e^{ipx/\hbar}$, $\alpha > 0$, $p \in \mathbb{R}$ ($\hbar = m = 1$ below).

1. *Normalisation:* $|\Psi(0,x)|^2 = \sqrt{2\alpha/\pi}\, e^{-2\alpha x^2}$, so $\int |\Psi|^2 dx = \sqrt{2\alpha/\pi}\sqrt{\pi/2\alpha} = 1$.
2. *Momentum space:* $\phi(p') = (2\pi\alpha)^{-1/4} e^{-(p'-p)^2/4\alpha}$, i.e. width $\Delta p = \hbar\sqrt{\alpha}$ and $\Delta x = 1/(2\sqrt{\alpha})$: a narrow packet in x is broad in p.
3. *Motion:* evaluating (1) gives $\Psi(t,x) = \dfrac{(2\alpha/\pi)^{1/4}}{\sqrt{1+2i\alpha t}} \exp\Big[-\dfrac{\alpha(x-pt)^2}{1+2i\alpha t} + ipx - \tfrac{i}{2}p^2 t \Big]$: the centre moves with velocity p/m and the packet spreads.

Numerics: `03_code/task1_gaussian_packets.py`.

3.2 Eigenfunctions in the continuum: not square integrable, degenerate

$$\hat{p}\, \tfrac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = p\, \tfrac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \qquad \hat{H}_0\, \tfrac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = \tfrac{p^2}{2m}\, \tfrac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}, \quad p \in \mathbb{R}.$$

These are not in L^2, the spectrum is **continuous**, $E = 0$ is non-degenerate and every $E > 0$ is **doubly degenerate** ($\pm p$: right- and left-going waves).

3.3 Bound spectrum: expansion in eigenstates

For a confining potential let $\hat{H}\varphi_j = E_j \varphi_j$, $j = 0, 1, 2, \ldots$ be a complete orthonormal basis of $L^2(\mathbb{R})$. With $\Psi(0,x) = \Psi_0(x) = \sum_j c_j \varphi_j(x)$,

$$\Psi(t,x) = \sum_j c_j \varphi_j(x) e^{-iE_j t/\hbar}, \qquad c_j = \int dx\, \varphi_j^*(x) \Psi_0(x) = \langle \varphi_j | \Psi_0 \rangle,$$

where the coefficient is extracted using $\langle \varphi_j | \varphi_{j'} \rangle = \delta_{jj'}$ (Kronecker delta).

3.4 Dirac delta as the continuous analogue of $\delta_{jj'}$

Inserting $\phi(p) = \int dx\, \bar{\psi}_p(x) \Psi_0(x)$ into the Fourier representation and exchanging the order of integration shows that consistency requires

$$\int_{-\infty}^{+\infty} \frac{dx}{2\pi\hbar}\, e^{-i(p-p')x/\hbar} = \delta(p-p'), \qquad \Phi(p) = \int dp'\, \Phi(p')\, \delta(p-p').$$

Definition — Dirac delta (sifting property) and its regularisation

$\delta(\xi)$ is characterised by $\int d\xi\, f(\xi)\delta(\xi) = f(0)$ for every "good" (smooth, decaying) test function f. It is not a function but a *distribution*. Regularise with a cut-off L (the regulator):

$$\delta_L(\xi) = \frac{1}{2\pi} \int_{-L}^{L} dx\, e^{i\xi x} = \frac{\sin L\xi}{\pi\xi} \quad \text{(Dirichlet kernel)}, \qquad \int d\xi f(\xi)\delta_L(\xi) \xrightarrow[L\to\infty]{} f(0).$$

Theorem (math): $\frac{1}{2\pi}\int_{-\infty}^{\infty} dx\, e^{i\xi x} = \delta(\xi)$.

Example — Task 2: numerical look at δ_L

$\delta_L(0) = L/\pi$ (height $\to \infty$, width $\sim \pi/L$, area 1); computed for $L = 10^1, \ldots, 10^5$ in `03_code/task2_dirac_delta.py`. The test function $f(\xi) = \xi e^{-10\xi^2}$ is odd and δ_L even, so $\int f\delta_L\, d\xi$ is *exactly* $0 = f(0)$ for every L; to see genuine convergence use an even-part test

function such as $(1+\xi)e^{-10\xi^2}$ (error $\sim e^{-L^2/40}$). Why $\delta_{mn} \to \delta(x-x')$: for a continuum, $\langle p|p'\rangle$ vanishes for $p \neq p'$, cannot equal 1 at $p = p'$ (the state is not normalisable), and only a unit-*area* spike reproduces the sifting property that δ_{mn} has for sums.

Key Formula Summary — Orthonormality and Completeness

	Discrete spectrum	Continuous spectrum
Orthonormality	$\int \varphi_m^* \varphi_n \, dx = \delta_{mn}$	$\int \psi_p^* \psi_{p'} \, dx = \delta(p-p')$
Completeness	$\sum_n \varphi_n(x)\varphi_n^*(x') = \delta(x-x')$	$\int dp \, \psi_p(x)\psi_p^*(x') = \delta(x-x')$
Operator form	$\sum_n \lvert n\rangle\langle n\rvert = \hat{1}$	$\int dp \, \lvert p\rangle\langle p\rvert = \hat{1}$

Any $f \in L^2$: $f = \sum_n c_n \varphi_n$ with $c_n = \langle \varphi_n | f \rangle$ (discrete) or $f(x) = \int dp \, \phi(p)\psi_p(x)$ with $\phi(p) = \langle \psi_p | f \rangle$ (continuous).

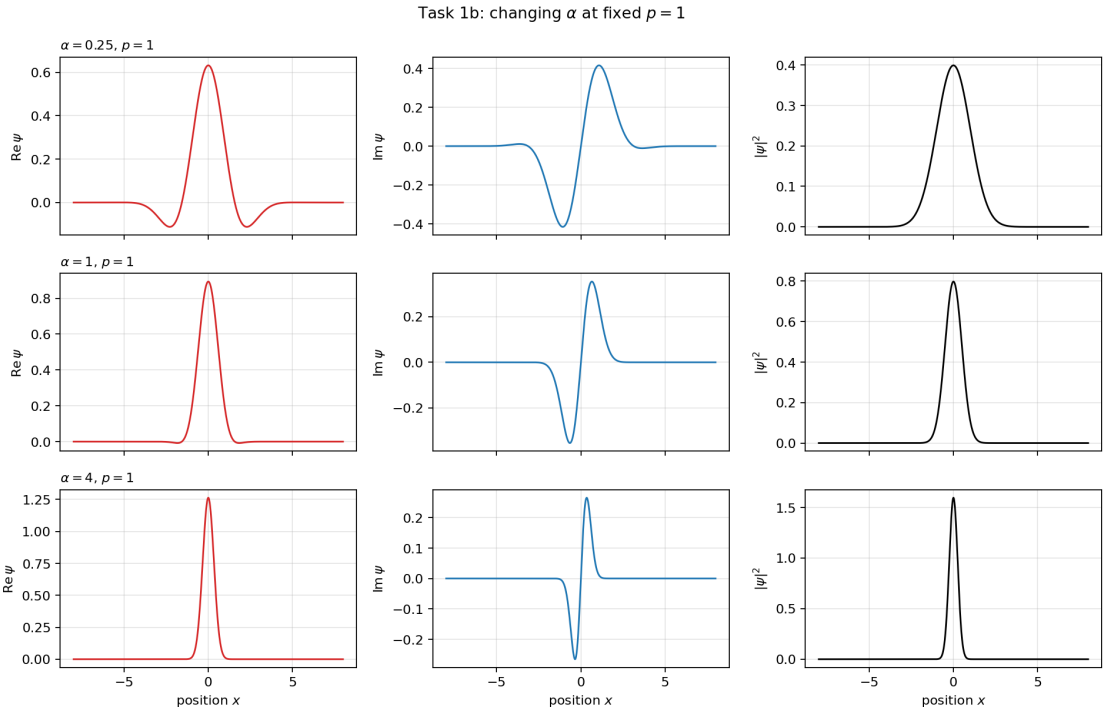
3.5 Project Task 1 — Free Gaussian Wavepackets

$\Psi(0,x) = (2\alpha/\pi)^{1/4} e^{-\alpha x^2} e^{ipx}$.

Task 1a — Unit normalisation

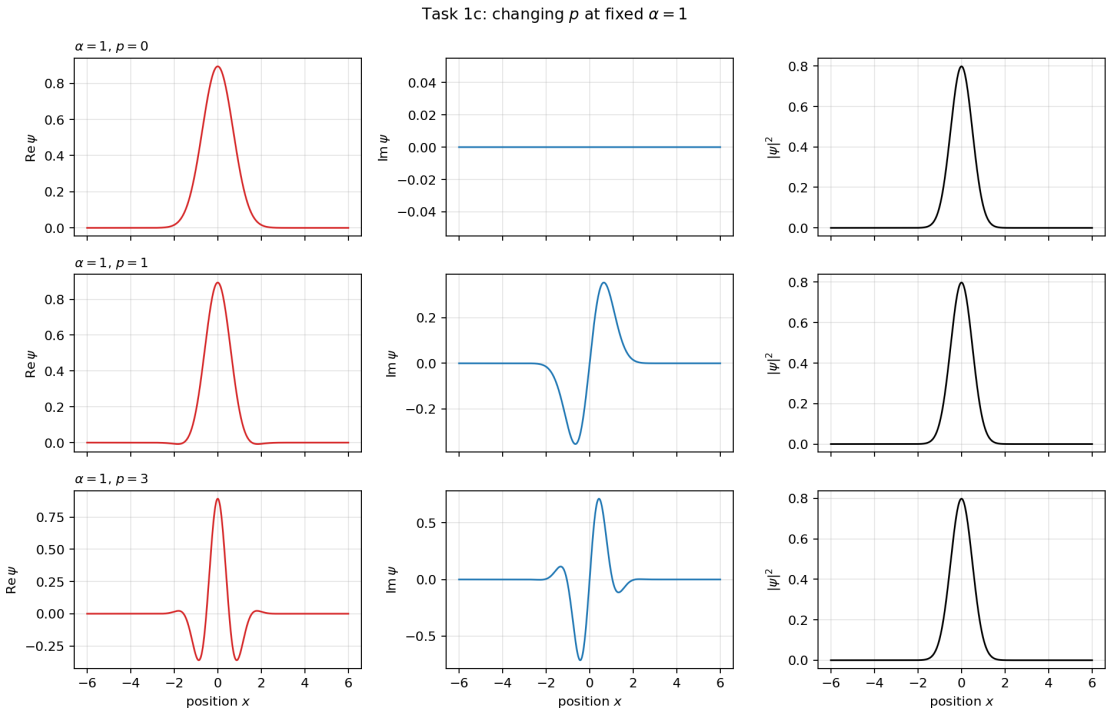
Analytically $\int |\Psi|^2 dx = \sqrt{2\alpha/\pi} \int e^{-2\alpha x^2} dx = 1$. Numerical quadrature for $\alpha \in \{0.1, 0.5, 1, 4, 10\}$ and $p \in \{0, 1, 5\}$ gave 1.000000000000 in all 15 cases; the momentum-space $\int |\Phi(p)|^2 dp$ is also 1.000000000000. The numerical Fourier evolution differs from the closed form by 6.6×10^{-16}.

Task 1b — Varying α at $p = 1$



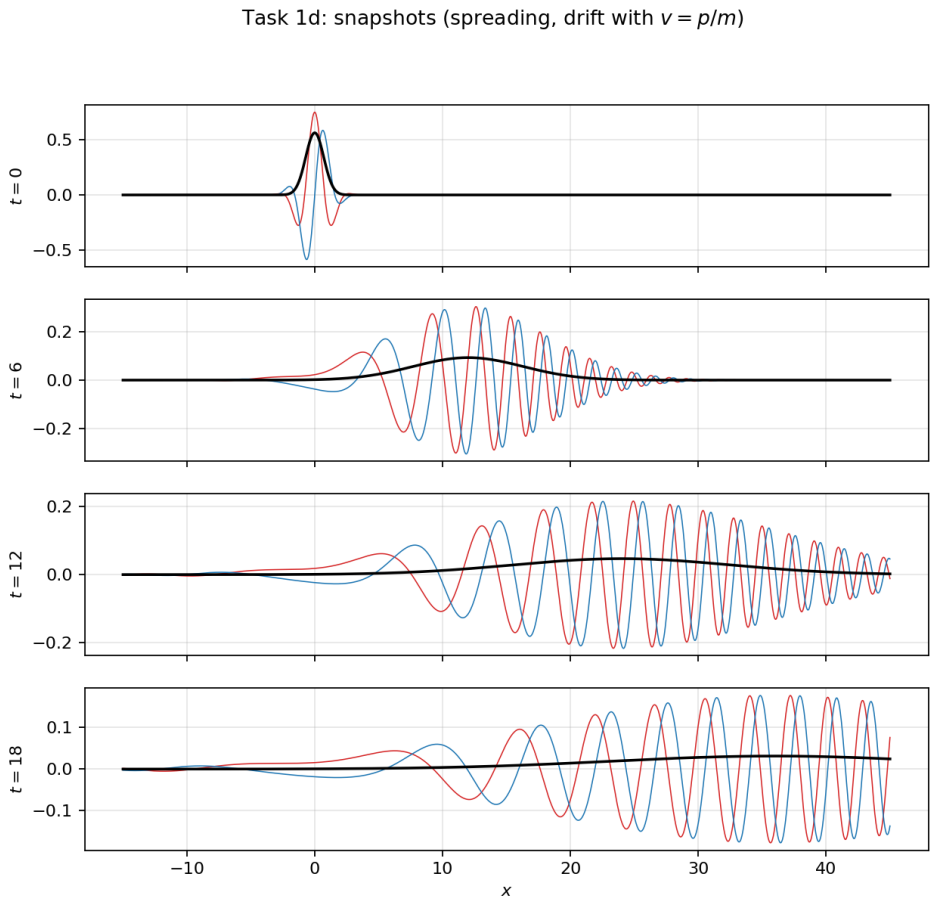
$\alpha = 0.25, 1, 4$: larger α gives a narrower $|\Psi|^2$ ($\Delta x = 1/2\sqrt{\alpha}$) with the same carrier oscillation e^{ix} (Re/Im oscillate with wavelength $2\pi/p$).

Task 1c — Varying p at $\alpha = 1$



$p = 0, 1, 3$: the envelope $|\Psi|^2$ is unchanged; only the oscillation frequency of $\mathrm{Re}\,\Psi$, $\mathrm{Im}\,\Psi$ increases. For $p = 0$, $\mathrm{Im}\,\Psi = 0$.

Task 1d — Animation



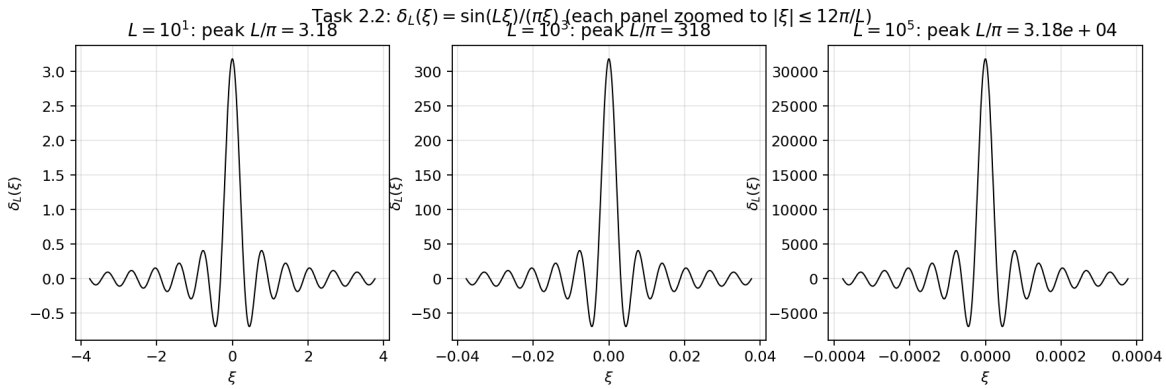
`animations/task1d_free_packet.gif` ($\alpha = 0.5$, $p = 2$): the centre moves as $x = pt$ and the packet spreads with width $\propto \sqrt{1 + (2\alpha t)^2}$.

3.6 Project Task 2 — Dirac Delta from the Dirichlet Kernel

Task 2.1 — $\delta_L(0)$

$\delta_L(0) = \frac{1}{2\pi}\int_{-L}^{L} dx = L/\pi$; the numerical integrals reproduce 3.183099, 31.83099, 318.3099, 3183.099, 31830.99 for $L = 10^1, \ldots, 10^5$: the peak height diverges.

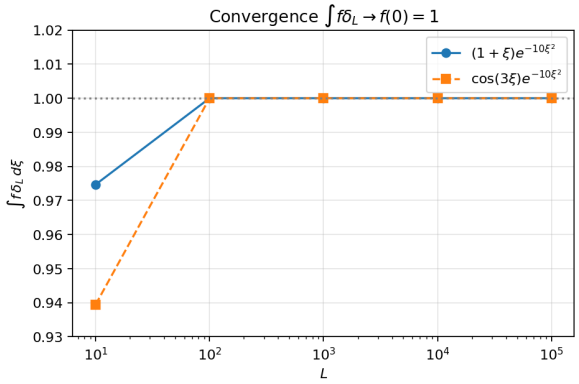
Task 2.2 — Graphs of $\sin(L\xi)/(\pi\xi)$



Task 2.3 — Test-function integrals

L	$f = \xi e^{-10\xi^2}$	$(1+\xi)e^{-10\xi^2}$	$\cos 3\xi\, e^{-10\xi^2}$
10	0	0.9746526813	0.9394123487
10^2	4×10^{-19}	1.0000000000	1.0000000000
10^3	-5×10^{-20}	1.0000000000	1.0000000000
10^4	9×10^{-18}	1.0000000000	1.0000000000
10^5	-8×10^{-18}	1.0000000000	1.0000000000
$f(0)$	0	1	1

The task's $f(\xi) = \xi e^{-10\xi^2}$ is odd while δ_L is even, so the integral is $0 = f(0)$ for every L (round-off level). Two even-part test functions show a non-trivial convergence to $f(0) = 1$, which is Gaussian-fast ($\sim e^{-L^2/40}$).



Task 2.4 — Why $\delta_{mn} \to \delta(x - x')$

For a discrete spectrum $\langle m|n\rangle = \delta_{mn}$ gives $c_m = \sum_n c_n \delta_{mn}$. In the continuum, the eigenfunctions $e^{ipx}/\sqrt{2\pi}$ cannot be normalised to 1 (not in L^2), but $\langle p|p'\rangle = \delta_L(p - p') \to \delta(p - p')$: it vanishes for $p \neq p'$ in the sense of distributions, diverges like L/π at $p = p'$, and has unit area for all L, so $\Phi(p) = \int \Phi(p')\delta(p - p')\, dp'$ plays the role of $c_m = \sum_n c_n \delta_{mn}$ (sum $\to$ integral, $\delta_{mn} \to \delta$).

4 Lecture 3 — Short-Ranged Potentials and Scattering Boundary Conditions

Definition — Short-ranged potential

V is **short-ranged** if $\lim_{x\to\pm\infty} V(x)\, x^m = 0$ for all $m = 0, 1, 2, \ldots$ (faster than any power). Example: $V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$ (the project potential) or $-V_0 e^{-\gamma x^2}$; but $-1/\sqrt{x^2 + 0.1}$ (a $1/|x|$ tail) is *not* short-ranged.

For $\hat{H} = \hat{p}^2/2m + V$ the spectrum has two parts:

- **bound states** $\hat{H}\varphi_j = E_j\varphi_j$, $E_j < 0$, $j = 0, \ldots, J$ ($J + 1$ levels, finite, possibly 0), square integrable;
- **continuum** $E > 0$, solutions of the TISE that are not square integrable, doubly degenerate.

4.1 Continuum solutions and scattering boundary conditions

For $x > x_R$ (beyond the range, $V = 0$) any solution is $Ae^{ikx} + Be^{-ikx}$ with $\hbar k = \sqrt{2mE} > 0$, A, B arbitrary complex; on the left similarly. Since $e^{\pm ikx} = e^{\pm ipx/\hbar}$ with $p = \hbar k$, the choices $(A, B) = (1, 0)$ or $(0, 1)$ label right- or left-moving asymptotics.

Definition — Scattering states $\psi_p(x)$, $T(p)$ and $R(p)$

For $p > 0$ (wave incident from the left, $E = p^2/2m$):

$$\psi_p(x \to +\infty) = T(p)\,\tfrac{1}{\sqrt{2\pi\hbar}} e^{+ipx/\hbar} \quad \text{(transmitted)},$$
$$\psi_p(x \to -\infty) = \tfrac{1}{\sqrt{2\pi\hbar}} e^{+ipx/\hbar} + R(p)\,\tfrac{1}{\sqrt{2\pi\hbar}} e^{-ipx/\hbar} \quad \text{(incident + reflected)}.$$

$T, R \in \mathbb{C}$ are the transmission/reflection coefficients; $|T|^2$, $|R|^2$ are the probabilities, and probability conservation reads $|T(p)|^2 + |R(p)|^2 = 1$. For $p < 0$ (incident from the right) the roles of the two sides swap. For every $E > 0$ there are therefore two degenerate scattering states.

4.2 Method: finite differences and the Neuhauser trick

Method — Finite-difference solution of the TISE as an initial-value problem

Discretise $x_m = x_0 + m\Delta x$ and replace $\psi''(x_m) \approx [\psi(x_{m+1}) - 2\psi(x_m) + \psi(x_{m-1})]/\Delta x^2$; the TISE $-\frac{\hbar^2}{2m}\psi'' + V\psi = E\psi$ becomes the three-term recurrence

$$\psi_{m-1} = 2\psi_m - \psi_{m+1} + \frac{2m\Delta x^2}{\hbar^2}\big(V(x_m) - E\big)\psi_m.$$

Neuhauser trick: start at large x with $\tilde{\psi}(x) = e^{ikx}$ (i.e. $A = 1$, $B = 0$, so only the transmitted wave exists), march to the left over the whole potential, and at very negative x fit $\tilde{\psi} = Ce^{ikx} + De^{-ikx}$. Then

$$\psi_p = \frac{1}{\sqrt{2\pi\hbar}}\frac{1}{C}\,\tilde{\psi}, \qquad T(p) = \frac{1}{C}, \qquad R(p) = \frac{D}{C}.$$

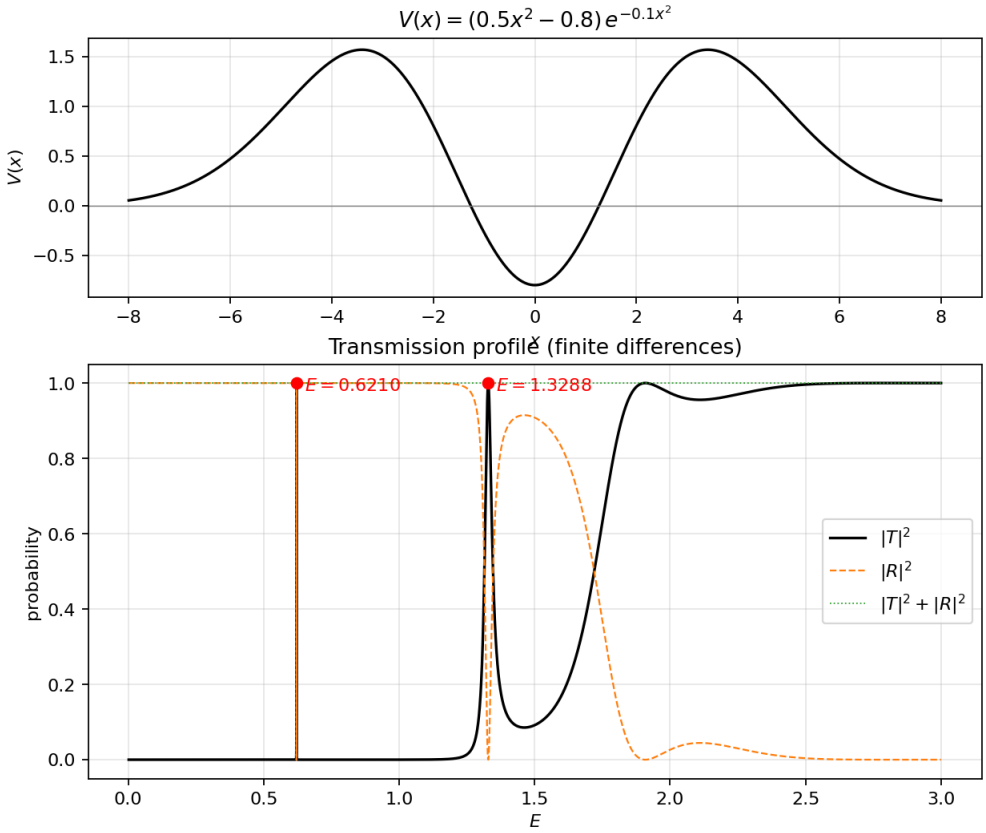
Example — Task 3: transmission profile and resonance tunnelling

For $V(x) = (0.5x^2 - 0.8)e^{-0.1x^2}$ (two barriers surrounding a well), $0 < E < 3$: compute T, R for each E, verify $|T|^2 + |R|^2 = 1$. The profile $|T(E)|^2$ shows two very sharp peaks at $E \simeq 0.6210$ and 1.3288 where $|T|^2 = 1$ although E is well below the barrier tops: **resonance tunnelling** ("resonance tunnelling diodes"), followed by broad overlapping features toward $|T|^2 \to 1$. Results: `04_results/figures/task3_*.png`, code `task3_transmission.py`.

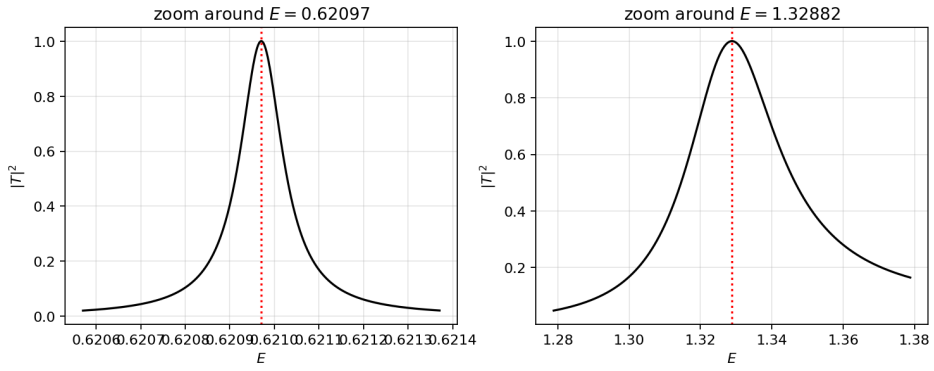
4.3 Project Task 3 — Transmission Profile

Method. Finite-difference recurrence with the Neuhauser trick (Lecture 3), $\Delta x = 0.002$, box $|x| \le 30$, lattice dispersion $\cos(k\Delta x) = 1 - E\Delta x^2$ for the plane waves; $T = 1/C$, $R = D/C$. Vectorised over all energies at once.

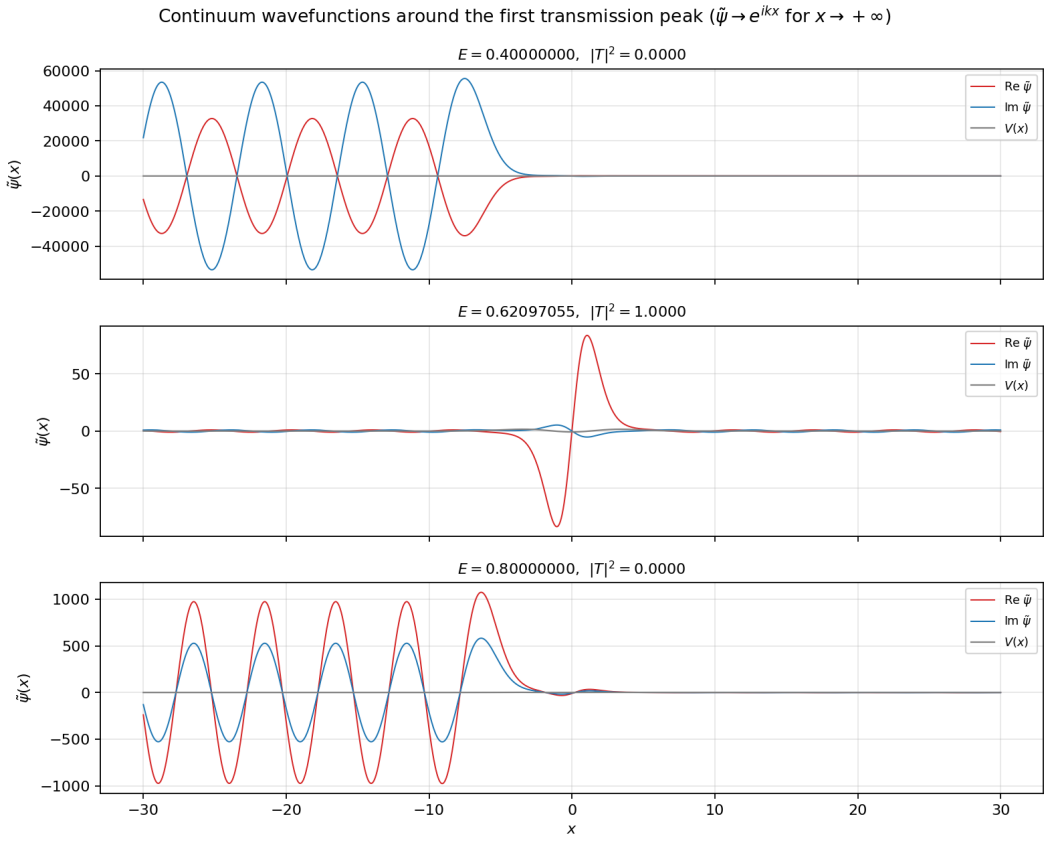
Results

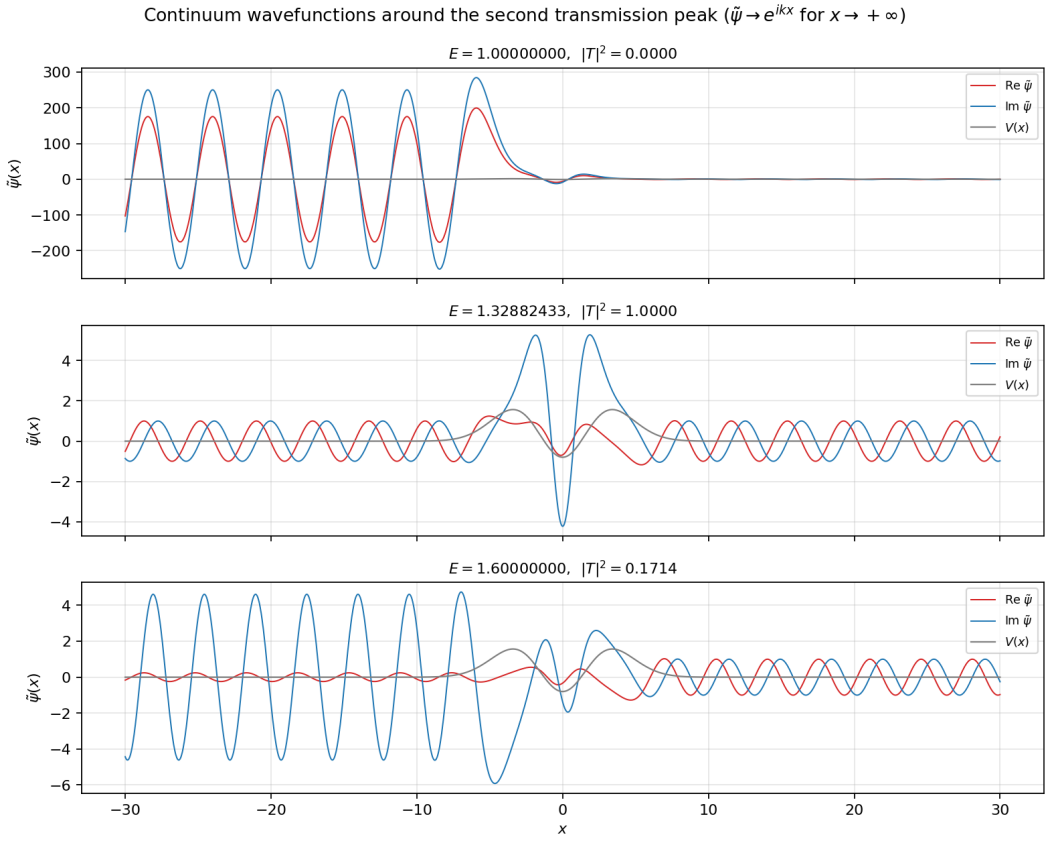


- Flux conservation: $\max_E \left||T|^2 + |R|^2 - 1\right| = 1.2 \times 10^{-10}$ on 6000 energies in $(0, 3)$.
- Peak 1: $E = 0.620971$, $|T|^2 = 1.0000$, FWHM $= 1.17 \times 10^{-4}$.
- Peak 2: $E = 1.328824$, $|T|^2 = 1.0000$, FWHM $= 3.2 \times 10^{-2}$.
- Cross-check against an adaptive Runge–Kutta solver (DOP853, rtol 10^{-12}): difference in $|T|^2$ is $\leq 6.5 \times 10^{-8}$ away from the narrow peak; *at* the $E = 0.620971$ peak it is 5.6×10^{-5} because the peak is $\sim 10^{-4}$ wide and the lattice and ODE peak positions differ by $O(10^{-6})$.
- The peaks agree with the Moodle plot (0.6210, 1.3288).



Continuum wavefunctions $\tilde{\psi}(x)$ ($\tilde{\psi} \to e^{ikx}$ for $x \to +\infty$)





On resonance $|\tilde{\psi}|$ is ≈ 1 outside but strongly enhanced inside the well (standing wave between the barriers), $|T|^2 = 1$. Off resonance the wave is almost completely reflected: $|C| \gg 1$, e.g.

$|C| \approx 1.5 \times 10^2$ at $E = 1.0$ ($|T|^2 = 4.3 \times 10^{-5}$), so $\tilde{\psi}$ is huge on the left.

5 Lectures 4–5 — Spectrum, Wavepackets, Asymptotics

Remark — Shape resonances

The narrow peaks correspond to states quasi-trapped in the well between two barriers; their energies $E_n = \varepsilon_n - \frac{\mathrm{i}}{2}\Gamma_n$ (anticipating L10) have small Γ_n (long lifetime). For $V = (0.5x^2 - 0.8)e^{-0.1x^2}$ the well supports one true bound state ($E_0 \approx -0.298$) and the resonance-like states at $\varepsilon \approx 0.6210$ and 1.33, which lie *above* $E = 0$ but below the barrier tops.

Theorem — Orthonormality and closure for a short-ranged potential

Let φ_j ($E_j < 0$) be the bound states and ψ_p ($p \in \mathbb{R} \setminus \{0\}$, both signs) the scattering states. Then

$$\int \varphi_j^* \varphi_{j'}\, dx = \delta_{jj'}, \quad \int \psi_p^* \psi_{p'}\, dx = \delta(p - p'), \quad \int \varphi_j^* \psi_p\, dx = 0,$$

$$\sum_j \varphi_j(x)\varphi_j^*(x') + \int_{-\infty}^{+\infty} dp\, \psi_p(x)\psi_p^*(x') = \delta(x - x').$$

Consequently any $f \in L^2$ is expanded as $f = \sum_j c_j\varphi_j + \int dp\, c(p)\psi_p$, and the TDSE is solved with $\Psi(t,x) = \sum_j c_j\varphi_j e^{-\mathrm{i}E_j t/\hbar} + \int dp\, \phi(p)\psi_p(x)e^{-\mathrm{i}\frac{p^2}{2m\hbar}t}$, $\phi(p) = \int dx\, \psi_p^*\Psi_0$. If Ψ_0 has no bound-state component then $c_j = 0$ for all j.

Example — Project part 4: wavepacket picture of a collision

Take an incoming Gaussian $\Psi(0,x)$ far to the left (momentum space $|\phi(p)|^2$ centred at p_0, width $\Delta p = \hbar\sqrt{\alpha}$, restricted to $[p_{\min}, p_{\max}] = [\sqrt{2mE_{\min}}, \sqrt{2mE_{\max}}]$), propagated by $\Psi(t,x) = \int dp\, \phi(p)\psi_p(x)e^{-\mathrm{i}E_p t/\hbar}$. Because the packet is narrow in p:

- if E_0 is off resonance ($T \approx 0$), the packet is reflected (*blocked*);
- if the packet is broader than the resonance width, it is *partially blocked*;
- if $E_0 = E_{\mathrm{res}}$ and the packet is *narrower* than Γ, it passes almost completely (*pass*), after being stored in the well for a time $\sim \hbar/\Gamma$.

Two solvers are used: Crank–Nicolson finite differences for the PDE ($(1 + \frac{\mathrm{i}\Delta t}{2\hbar}\hat{H})\Psi^{n+1} = (1 - \frac{\mathrm{i}\Delta t}{2\hbar}\hat{H})\Psi^n$, unitary, second order) and the ψ_p expansion above; they agree to $O(\Delta x^2)$.

Method — Asymptotic analysis

For large t the free integral $\int d\xi\, f(\xi)e^{-\mathrm{i}\Omega\xi}$ with smooth compactly supported f decays faster than any power of $1/\Omega$ (stationary-phase). This is the mathematical theorem behind the asymptotic condition below (Appendix III of the lecturer's notes).

5.1 Project Task 4 — Wavepacket Scattering (TDSE, Finite Differences)

Methods. (A) Crank–Nicolson finite differences for $i\partial_t\Psi = [-\frac{1}{2}\partial_{xx} + V]\Psi$ (unitary, second order in $\Delta x, \Delta t$; norm conserved to 10^{-6}); (B) expansion in scattering states (Lecture 4), $\Psi(t,x) = \int dp\, \phi(p)\psi_p^+(x)e^{-ip^2t/2}$, where $\psi_p^+ = T\tilde{\psi}/\sqrt{2\pi}$ inside $|x| < 25$ (finite-difference solution) and analytic

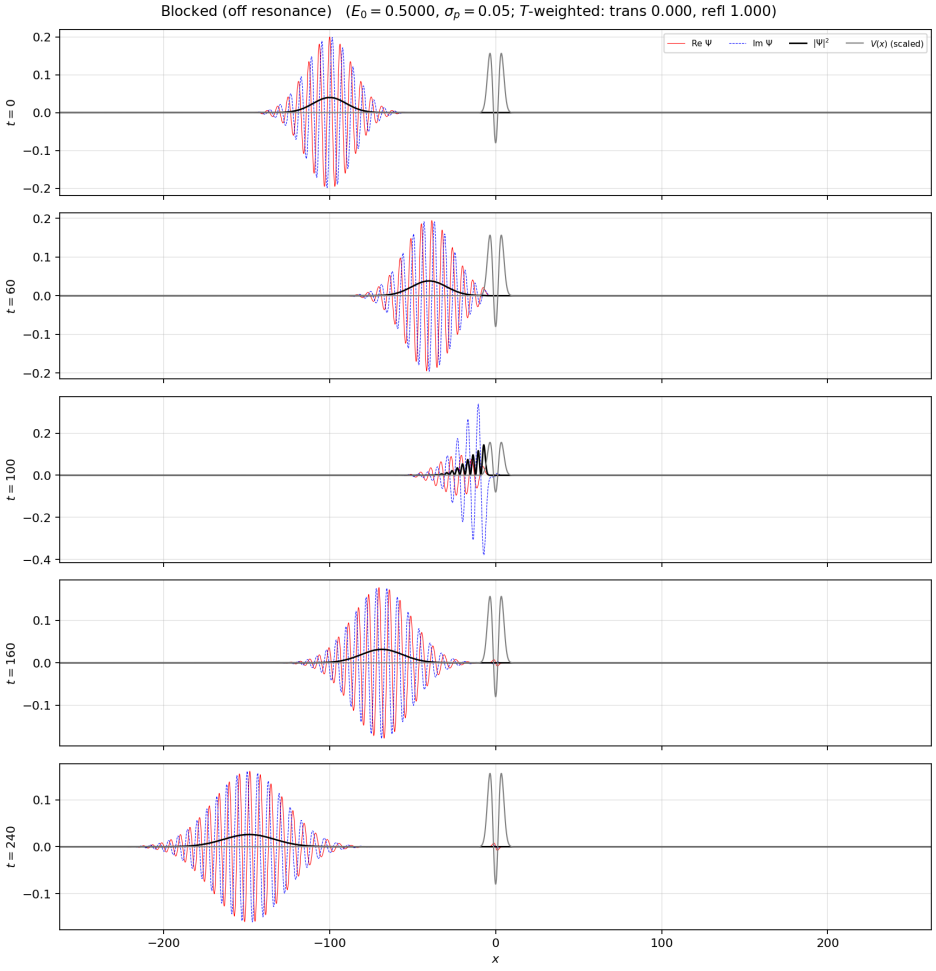
plane waves outside. Initial state $\Psi(0,x) = (2a/\pi)^{1/4} e^{-a(x-x_0)^2 + ip_0(x-x_0)}$, $a = \sigma_p^2$. Method B is needed for the first (very narrow) resonance where a grid would need $\sim 10^6$ points.

Remark — Cross-validation of the two methods

For the broad packet ($E_0 = 1.3288$, $\sigma_p = 0.05$) the maximum difference $\max_x \left||\Psi|^2_{CN} - |\Psi|^2_{\mathrm{B}}\right|$ at $t = 100$ falls with resolution as $1.6 \times 10^{-3} \to 4.0 \times 10^{-4} \to 3.9 \times 10^{-5}$ for $(\Delta x, \Delta t) = (0.05, 0.1), (0.025, 0.05), (0.0125, 0.0125)$ (peak $|\Psi|^2 \approx 3.9 \times 10^{-2}$): $O(h^2)$ convergence of the CN scheme toward the exact expansion. At $t = 240$: $2.1 \times 10^{-3} \to 5.1 \times 10^{-4} \to 5.1 \times 10^{-5}$.

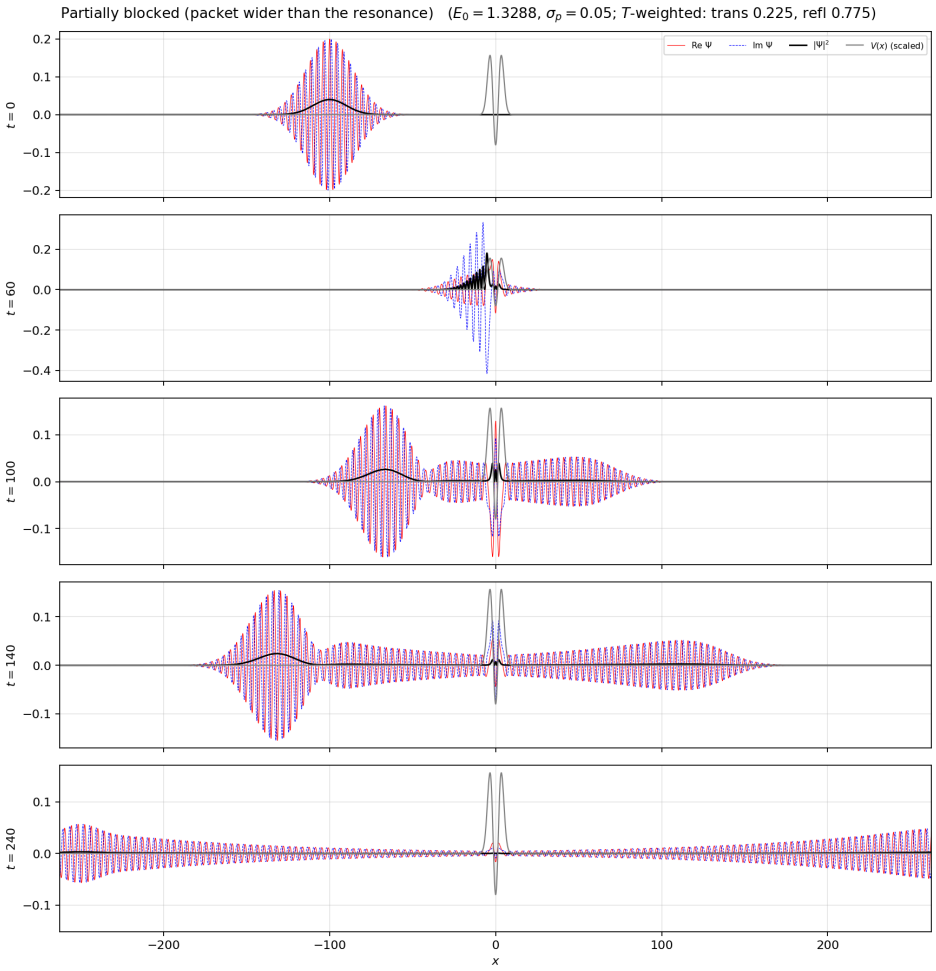
Task 4a — Blocked (off resonance)

$E_0 = 0.5$, $\sigma_p = 0.05$. Transmitted probability $P_{\mathrm{trans}} = 1.0 \times 10^{-4}$ (CN), theory $\int |\phi|^2 |T|^2 = 1 \times 10^{-5}$; reflected 0.9999.



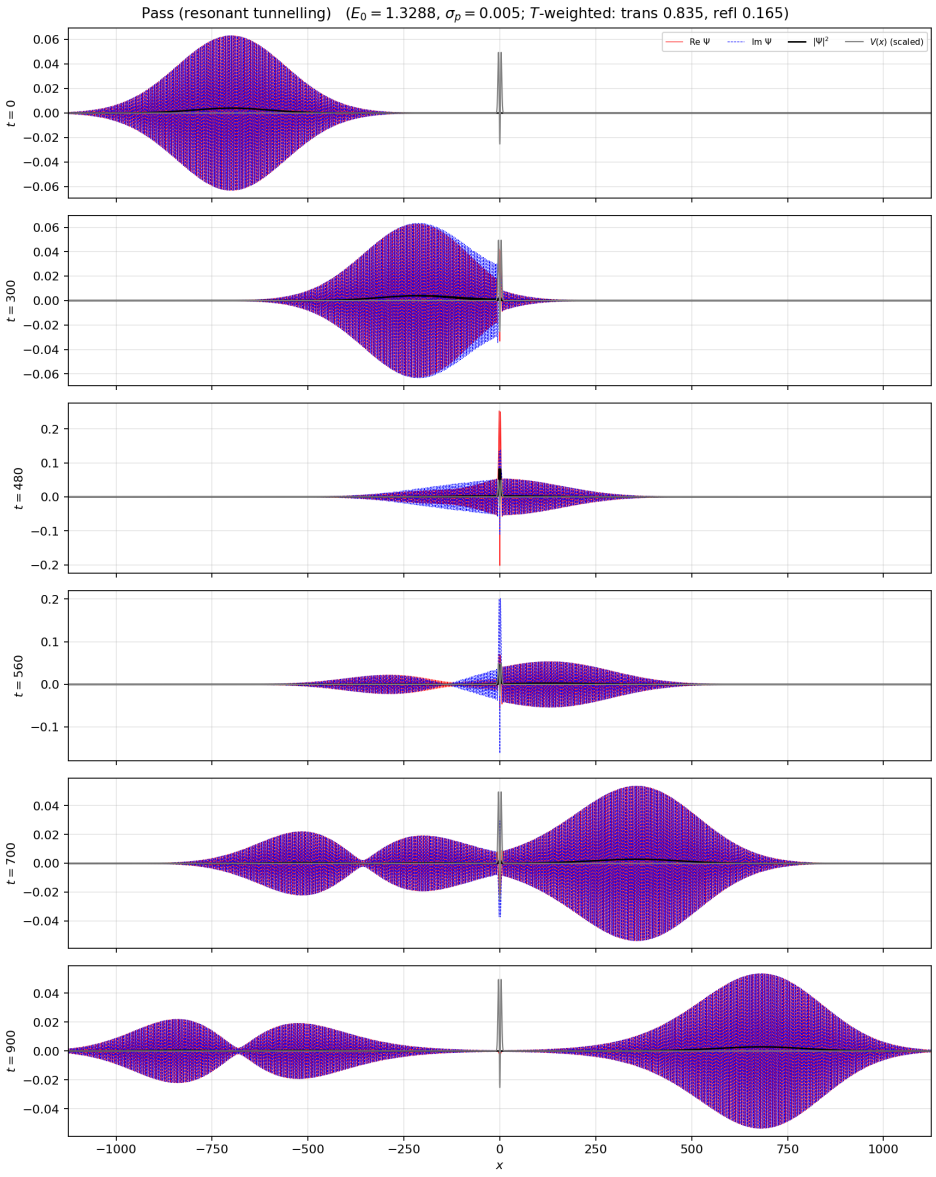
Task 4b — Partially blocked

Second resonance, $E_0 = 1.3288$, $\sigma_p = 0.05$ (energy spread $0.08 \gg \Gamma_2 = 0.031$): $P_{\mathrm{trans}} = 0.2246$ (CN) vs 0.2252 (theory).



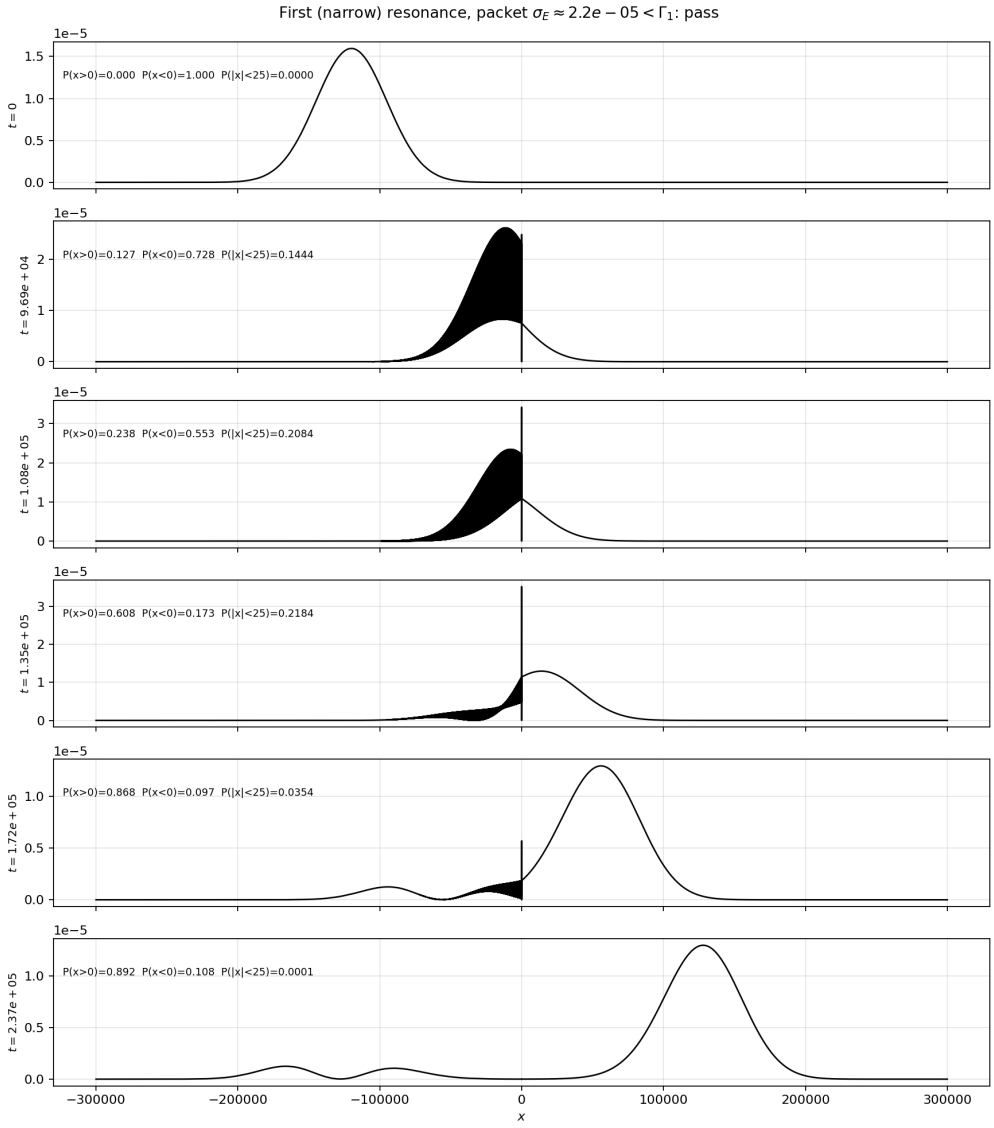
Task 4c — Pass through the second resonance

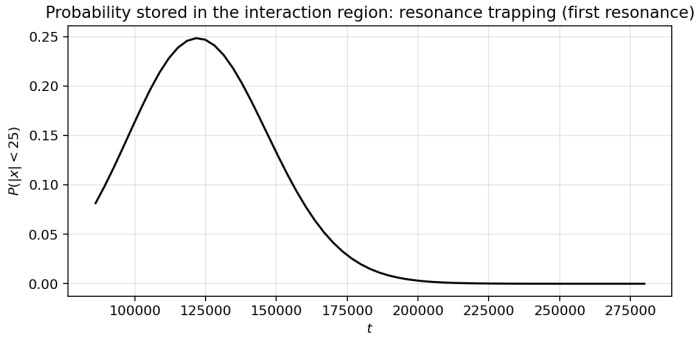
$E_0 = 1.3288$, $\sigma_p = 0.005$ (energy spread $0.008 < \Gamma_2$), $x_0 = -700$: $P_{\text{trans}} = 0.833$ (CN) vs 0.835 (theory); the packet is delayed inside the well and leaves as a transmitted plus a smaller reflected packet.



Task 4d — First (narrow) resonance, Method B

$E_0 = 0.620971$, $\sigma_p = 2 \times 10^{-5}$ ($\sigma_E = 2.2 \times 10^{-5} < \Gamma_1 = 1.2 \times 10^{-4}$), $x_0 = -1.2 \times 10^5$. Probability bookkeeping: $P_{\text{trans}} \to 0.892$, $P_{\text{refl}} \to 0.108$ (theory 0.8923/0.1077); up to 24.8% of the probability is stored in $|x| < 25$ at $t \approx 1.2 \times 10^5$; the packet returns to full free motion with an enormous delay $\sim 1/\Gamma_1 \sim 8.6 \times 10^3$. Detuned to $E_0 = 0.60$ the same packet is blocked ($P_{\text{refl}} = 1.0000$).





Black bands are real interference fringes (incident + reflected waves) sampled at $\Delta x = 5$. Animations: `animations/task4_*.gif`.

6 Lectures 5–6 — Formal Scattering Theory in the Time Domain

Setting: $\hat{H} = \hat{H}_0 + \hat{V}$, $\hat{H}_0 = \hat{p}^2/2m$, V short-ranged. To begin, assume there are *no bound states* (or that $\Psi(0)$ does not populate them).

Theorem — Asymptotic condition

For every $|\Psi(0)\rangle$ orthogonal to all bound states there exist unique $|\phi_{\text{in}}\rangle$, $|\phi_{\text{out}}\rangle$ with

$$e^{-\mathrm{i}\hat{H}t/\hbar}|\Psi(0)\rangle - e^{-\mathrm{i}\hat{H}_0 t/\hbar}|\phi_{\text{in}}\rangle \xrightarrow[t\to-\infty]{} 0, \qquad e^{-\mathrm{i}\hat{H}t/\hbar}|\Psi(0)\rangle - e^{-\mathrm{i}\hat{H}_0 t/\hbar}|\phi_{\text{out}}\rangle \xrightarrow[t\to+\infty]{} 0.$$

Interacting motion looks free before and after the collision. The basic question of scattering theory is the map

$$|\phi_{\text{out}}\rangle = \hat{S}\,|\phi_{\text{in}}\rangle \qquad \text{(scattering operator)}.$$

6.1 Construction of $\hat{S}$

From the two limits of the asymptotic condition,

$$|\Psi(0)\rangle = \lim_{T\to\infty} e^{\mathrm{i}\hat{H}T/\hbar}e^{-\mathrm{i}\hat{H}_0 T/\hbar}|\phi_{\text{in}}\rangle =: \hat{\Omega}_+|\phi_{\text{in}}\rangle, \quad |\phi_{\text{out}}\rangle = \lim_{T\to\infty} e^{\mathrm{i}\hat{H}_0 T/\hbar}e^{-\mathrm{i}\hat{H}T/\hbar}|\Psi(0)\rangle.$$

Hence

$$\hat{S} = \lim_{T\to\infty} e^{\mathrm{i}\hat{H}_0 T/\hbar}e^{-2\mathrm{i}\hat{H}T/\hbar}e^{\mathrm{i}\hat{H}_0 T/\hbar}, \qquad \hat{S}^\dagger = \hat{S}^{-1}.$$

Warning — Operator exponentials do not combine

Since $\hat{H}_0$ and $\hat{V}$ do not commute, $e^A e^B \neq e^{A+B}$, so $e^{-\mathrm{i}\hat{H}_0 t}e^{\mathrm{i}\hat{H}t} \neq e^{-\mathrm{i}(\hat{H}_0-\hat{H})t}$. The limit above is a *strong* limit, $\lim_\lambda \hat{A}(\lambda)|\psi\rangle = \hat{A}_\infty|\psi\rangle$; weak convergence only controls $\langle\psi_1|\hat{A}(\lambda)|\psi_2\rangle$ (Taylor, *Scattering Theory*).

Proof — Integral (Duhamel) form of $|\Psi\rangle$

Since $\frac{d}{dt}\big(e^{\mathrm{i}\hat{H}t/\hbar}e^{-\mathrm{i}\hat{H}_0 t/\hbar}\big) = \frac{\mathrm{i}}{\hbar}e^{\mathrm{i}\hat{H}t/\hbar}(\hat{H}-\hat{H}_0)e^{-\mathrm{i}\hat{H}_0 t/\hbar}$ and $\hat{H}-\hat{H}_0 = \hat{V} = V(\hat{x})$,

$$|\Psi\rangle = |\phi_{\text{in}}\rangle - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{0} dt\; e^{\mathrm{i}\hat{H}t/\hbar}\,\hat{V}\,e^{-\mathrm{i}\hat{H}_0 t/\hbar}|\phi_{\text{in}}\rangle.$$

(Integrate the derivative from $t\to-\infty$ to 0 and use $\hat{\Omega}(0)=1$.) ■

6.2 The iε prescription and the Lippmann–Schwinger equation

Expand $|\phi_{\text{in}}\rangle = \int dp\,\Phi_{\text{in}}(p)|\psi_p^{\text{F}}\rangle$, $\Phi_{\text{in}}(p) = \langle\psi_p^{\text{F}}|\phi_{\text{in}}\rangle$, and use $e^{-\mathrm{i}\hat{H}_0 t/\hbar}|\psi_p^{\text{F}}\rangle = e^{-\mathrm{i}E_p t/\hbar}|\psi_p^{\text{F}}\rangle$. Insert a convergence factor $e^{\varepsilon t/\hbar}$, $\varepsilon\to 0^+$ (legitimate since $e^{\varepsilon t}\to 1$), which makes the time integral converge at $t\to-\infty$:

$$\int_{-\infty}^{0} dt\, e^{-\frac{\mathrm{i}}{\hbar}(E_p-\hat{H}+\mathrm{i}\varepsilon)t} = \frac{\mathrm{i}\hbar}{E_p-\hat{H}+\mathrm{i}\varepsilon}.$$

Definition — Green operators and Lippmann–Schwinger states

$$\hat{G}^{\pm}(E) = \frac{1}{E-\hat{H}\pm\mathrm{i}\varepsilon} \quad \text{(retarded/advanced)}, \qquad |\psi_p^{\pm}\rangle = |\psi_p^{\text{F}}\rangle + \frac{1}{E_p-\hat{H}\pm\mathrm{i}\varepsilon}\hat{V}|\psi_p^{\text{F}}\rangle$$

(explicit form of the LSE). Equivalently, with the free resolvent (*implicit form*)

$$|\psi_p^{\pm}\rangle = |\psi_p^{\text{F}}\rangle + \frac{1}{E_p-\hat{H}_0\pm\mathrm{i}\varepsilon}\hat{V}|\psi_p^{\pm}\rangle.$$

Position representation: $G_E^{\pm}(x,x') = \langle x|(E-\hat{H}\pm\mathrm{i}\varepsilon)^{-1}|x'\rangle$.

The results: $|\Psi\rangle = \int dp\,\Phi_{\rm in}(p)|\psi_p^+\rangle = \int dp\,\Phi_{\rm out}(p)|\psi_p^-\rangle$, and $|\phi_{\rm out}\rangle = \int dp\,\Phi_{\rm out}(p)|\psi_p^{\rm F}\rangle$: the ψ^+ (resp. ψ^-) are the Møller-transformed free states that connect to $\phi_{\rm in}$ (resp. $\phi_{\rm out}$).

Remark — Digression: Cauchy principal value

For a test function f, $\int dx\,f(x)/x$ diverges at 0; $\mathrm{P}\int dx\,f(x)/x = \lim_{\varepsilon\to 0^+}\left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)$. With $\frac{1}{x\pm \mathrm{i}\varepsilon} = \frac{x\mp \mathrm{i}\varepsilon}{x^2+\varepsilon^2}$ and the Lorentzian $f_\varepsilon(x) = \frac{1}{\pi}\frac{\varepsilon}{x^2+\varepsilon^2} \to \delta(x)$ (unit area for every ε),

$$\frac{1}{x \pm \mathrm{i}\varepsilon} = \mathrm{P}\frac{1}{x} \mp \mathrm{i}\pi\,\delta(x), \qquad \frac{1}{x+\mathrm{i}\varepsilon} - \frac{1}{x-\mathrm{i}\varepsilon} = -2\pi\mathrm{i}\,\delta(x).$$

7 Lecture 7 — Green Functions, Born Series, and $T(p)$, $R(p)$ from the LSE

Proposition — Consequences of the LSE

(1) $(E-\hat H+\mathrm{i}\varepsilon)|\psi_p^+\rangle = (E-\hat H_0)|\psi_p^{\rm F}\rangle + \hat V|\psi_p^{\rm F}\rangle = 0$ for $E=E_p$, i.e. $\hat H|\psi_p^+\rangle = E_p|\psi_p^+\rangle$: $\psi_p^\pm$ are genuine eigenstates of the full $\hat H$.
(2) Iterating the implicit form gives the **Born series**

$$|\psi_p^+\rangle = |\psi_p^{\rm F}\rangle + \hat G_0^+\hat V|\psi_p^{\rm F}\rangle + \hat G_0^+\hat V\hat G_0^+\hat V|\psi_p^{\rm F}\rangle + \cdots, \quad \hat G_0^+ = \frac{1}{E-\hat H_0+\mathrm{i}\varepsilon},$$

useful when V is weak (first Born approximation: stop after one $\hat V$).

Definition — Free retarded Green function in 1D

Using $\hat H_0 = \int dp'|\psi_{p'}^{\rm F}\rangle\frac{p'^2}{2m}\langle\psi_{p'}^{\rm F}|$,

$$G_{E,0}^\pm(x,x') = \int dp' \frac{\psi_{p'}^{\rm F}(x)\psi_{p'}^{\rm F*}(x')}{E-\frac{p'^2}{2m}\pm\mathrm{i}\varepsilon} = \mp\mathrm{i}\frac{m}{\hbar p}\,e^{\pm \mathrm{i}p|x-x'|/\hbar}, \qquad p=\sqrt{2mE}.$$

Theorem — $T(p)$ and $R(p)$ as matrix elements of $\hat V$

Insert $G_{E,0}^+$ in the coordinate form of the LSE, $\psi_p^+(x) = \psi_p^{\rm F}(x) + \int dy\,G_{E,0}^+(x,y)V(y)\psi_p^+(y)$, and let $x\to\pm\infty$ (with V supported in $[a,b]$). For $p>0$:

$$T(p) = 1 - \frac{\mathrm{i}m}{p}\sqrt{2\pi\hbar}\int_a^b dy\,e^{-\mathrm{i}py/\hbar}V(y)\psi_p^+(y) = 1 - 2\pi\mathrm{i}\frac{m}{p}\,\langle\psi_p^{\rm F}|\hat V|\psi_p^+\rangle \ (\hbar=1),$$
$$R(p) = -\frac{\mathrm{i}m}{p}\sqrt{2\pi\hbar}\int_a^b dy\,e^{+\mathrm{i}py/\hbar}V(y)\psi_p^+(y) = -2\pi\mathrm{i}\frac{m}{|p|}\,\langle\psi_{-p}^{\rm F}|\hat V|\psi_p^+\rangle \ (\hbar=1).$$

8 Lecture 8 — The $\hat{S}$ Matrix, the $\hat{T}$ Operator, Energy Conservation

Theorem — $\hat{S}$ in the scattering basis

With $|\phi_{\text{out}}\rangle = \int dp\, C_{\text{out}}(p)|\psi_p^{\text{F}}\rangle$, $C_{\text{out}}(p) = \langle\psi_p^-|\Psi\rangle$, and $C_{\text{in}}(p') = \langle\psi_{p'}^{\text{F}}|\phi_{\text{in}}\rangle$,

$$\hat{S} = \int dp \int dp'\, |\psi_p^{\text{F}}\rangle\, \langle\psi_p^-|\psi_{p'}^+\rangle\, \langle\psi_{p'}^{\text{F}}|, \qquad \langle\psi_p^{\text{F}}|\hat{S}|\psi_{p'}^{\text{F}}\rangle = \langle\psi_p^-|\psi_{p'}^+\rangle.$$

Proof — Evaluation of $\langle\psi_p^-|\psi_{p'}^+\rangle$

Use the implicit LSE for $\psi_{p'}^+$ and the conjugate of the explicit one for $\langle\psi_p^-|$, together with $\hat{H}|\psi_{p'}^+\rangle = E_{p'}|\psi_{p'}^+\rangle$ and $\langle\psi_p^{\text{F}}|\hat{H}_0 = E_p\langle\psi_p^{\text{F}}|$:

$$\begin{aligned}\langle\psi_p^-|\psi_{p'}^+\rangle &= \delta(p-p') + \langle\psi_p^{\text{F}}|\hat{V}|\psi_{p'}^+\rangle\left[\tfrac{1}{E_{p'}-E_p+\mathrm{i}\varepsilon} + \tfrac{1}{E_p-E_{p'}+\mathrm{i}\varepsilon}\right] \\ &= \delta(p-p') - 2\pi\mathrm{i}\,\delta(E_p-E_{p'})\,\langle\psi_p^{\text{F}}|\hat{V}|\psi_{p'}^+\rangle,\end{aligned}$$

where $\frac{1}{u+\mathrm{i}\varepsilon} - \frac{1}{u-\mathrm{i}\varepsilon} = -2\pi\mathrm{i}\delta(u)$ was used. ■

Definition — $\hat{T}$ operator and on-shell T-matrix

$$\hat{T}(E) = \hat{V} + \hat{V}\frac{1}{E-\hat{H}+\mathrm{i}\varepsilon}\hat{V}, \qquad \hat{T}|\psi_p^{\text{F}}\rangle = \hat{V}|\psi_p^+\rangle, \qquad \hat{T} = \hat{V} + \hat{V}\hat{G}_0^+\hat{V} + \hat{V}\hat{G}_0^+\hat{V}\hat{G}_0^+\hat{V} + \cdots$$

(*Born expansion for* $\hat{T}$). It carries all scattering information:

$$\boxed{\langle\psi_p^{\text{F}}|\hat{S}|\psi_{p'}^{\text{F}}\rangle = \delta(p-p') - 2\pi\mathrm{i}\,\delta(E_p-E_{p'})\,\langle\psi_p^{\text{F}}|\hat{T}(E_{p'})|\psi_{p'}^{\text{F}}\rangle.}$$

Remark — Energy conservation and on-shell

The factor $\delta(E_p - E_{p'})$ means the S-matrix element vanishes unless $E_p = E_{p'}$: *scattering off a static potential is elastic.* Only the **on-shell** matrix elements ($E_p = E_{p'}$) of $\hat{T}$ matter.

9 Lecture 9 — Transmission and Reflection Probabilities; Variational Bound States

9.1 From $\hat{S}$ to T and R

Apply $\hat{S}$ to $\phi_{\text{in}}(p)$ and use $\delta(E_p - E_{p'}) = \frac{m}{|p|}\big[\delta(p-p') + \delta(p+p')\big]$ (from $\delta(F(\xi)) = \sum_\nu \frac{1}{|F'(\xi_\nu)|}\delta(\xi-\xi_\nu)$ and $\delta(a\xi) = \frac{1}{|a|}\delta(\xi)$). With $p \mapsto -p$ in the second term,

$$|\phi_{\text{out}}\rangle = \underbrace{\int dp\, |\psi_p^{\text{F}}\rangle \Phi_{\text{in}}(p)\, \mathcal{T}(p)}_{|\phi_{\text{trans}}\rangle} + \underbrace{\int_{p_{\min}}^{p_{\max}} dp\, |\psi_{-p}^{\text{F}}\rangle \Phi_{\text{in}}(p)\, \mathcal{R}(p)}_{|\phi_{\text{refl}}\rangle},$$

where $\mathcal{T}(p) = 1 - 2\pi\mathrm{i}\frac{m}{|p|}\langle\psi_p^{\text{F}}|\hat{V}|\psi_p^+\rangle = T(p)$ and $\mathcal{R}(p) = -2\pi\mathrm{i}\frac{m}{|p|}\langle\psi_{-p}^{\text{F}}|\hat{V}|\psi_p^+\rangle = R(p)$, identical to Lecture 7. The wavepacket is thus split into a transmitted and a reflected square-integrable packet

with $\langle\phi_{\text{trans}}|\phi_{\text{refl}}\rangle = 0$ and

$$\langle\phi_{\text{trans}}|\phi_{\text{trans}}\rangle = \int dp\,|\Phi_{\text{in}}(p)T(p)|^2, \quad \langle\phi_{\text{refl}}|\phi_{\text{refl}}\rangle = \int dp\,|\Phi_{\text{in}}(p)R(p)|^2, \quad \text{sum} = \langle\phi_{\text{out}}|\phi_{\text{out}}\rangle = 1.$$

Corollary — Narrow packets reproduce the stationary T and R

Choose a packet so narrow in p that $T(p) \approx T(p_0)$, $R(p) \approx R(p_0)$ over its support. Then

$$P_{\text{trans}} = |T(p_0)|^2, \qquad P_{\text{refl}} = |R(p_0)|^2, \qquad |T|^2 + |R|^2 = 1.$$

For resonance peaks "narrow enough" means $\Delta E \ll \Gamma$; otherwise the packet is only partially transmitted.

9.2 Project part 5 — Variational (basis-set) method for bound states

Method — Variational calculation in a complete orthonormal basis

Expand $\psi_n(x) = \sum_{j=1}^{J} c_{jn}\varphi_j(x)$ in a basis $\{\varphi_j\}$. Then $\hat{H}\psi_n = E_n\psi_n$ becomes the matrix eigenproblem

$$\underline{\mathbf{H}}\,\vec{c}^{(n)} = E_n\vec{c}^{(n)}, \qquad H_{jj'} = \langle\varphi_j|\hat{H}|\varphi_{j'}\rangle,$$

truncated at J (e.g. $J = 100, 200, \ldots$); eigenvalues are ordered $E_1 \le \cdots \le E_J$ and $\vec{c}^{(n)*}\cdot\vec{c}^{(n)} = 1$. Use the **particle-in-a-box basis** on $(-L/2, L/2)$: $\varphi_j(x) = \sqrt{2/L}\sin\big(j\pi(x + L/2)/L\big)$, so

$$H_{jj'} = \frac{\hbar^2\pi^2 j^2}{2mL^2}\delta_{jj'} + \int_{-L/2}^{L/2} \varphi_j V \varphi_{j'}\,dx.$$

Test: $V = \frac{1}{2}x^2 - 0.8$ gives $E_n = n + \frac{1}{2} - 0.8$ exactly (harmonic oscillator, $\hbar = m = \omega = 1$).

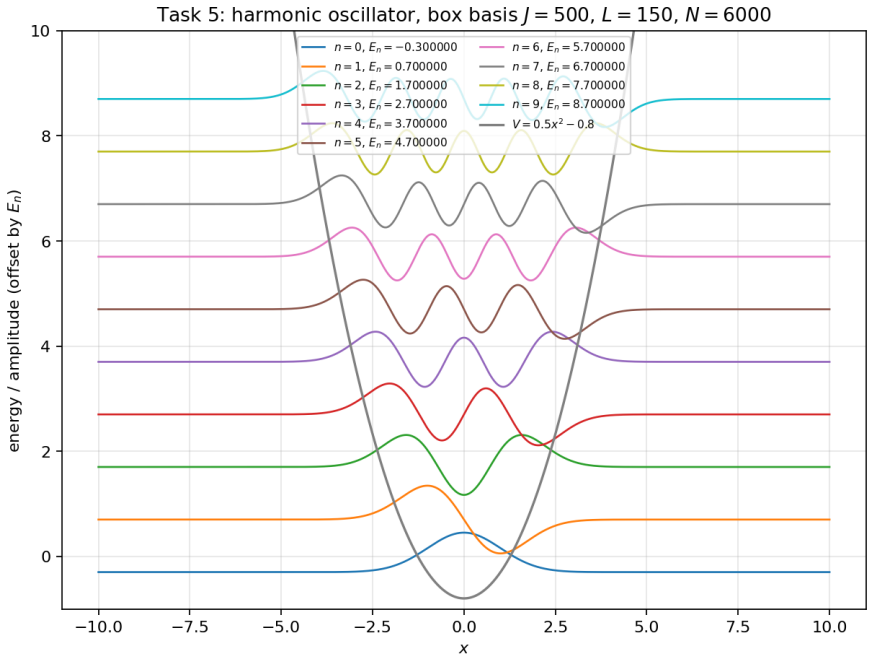
For the short-ranged $V = (0.5x^2 - 0.8)e^{-0.1x^2}$ this yields one true bound state $E_0 = -0.29796$; the remaining box levels approximate the continuum, but two *localised* levels near 0.6210 and 1.33 mimic the resonances. A box of size L only discretises the continuum, so localisation must be tested by the weight $\int_{|x|<10}|\psi_n|^2dx$ (Task 5).

9.3 Project Task 5 — Variational Bound States

Method. Box basis $\varphi_j = \sqrt{2/L}\sin(j\pi(x+L/2)/L)$, $H_{jj'} = \frac{(j\pi/L)^2}{2}\delta_{jj'} + V_{jj'}$ with $V_{jj'}$ by trapezoid quadrature, dense symmetric eigensolver.

Task 5a — Harmonic oscillator $V = 0.5x^2 - 0.8$

$J = 500$, $L = 150$, $N = 6000$ quadrature points. Result: $E_n = n - 0.3$ to 10^{-12} for $n = 0, \ldots, 9$ (exact).



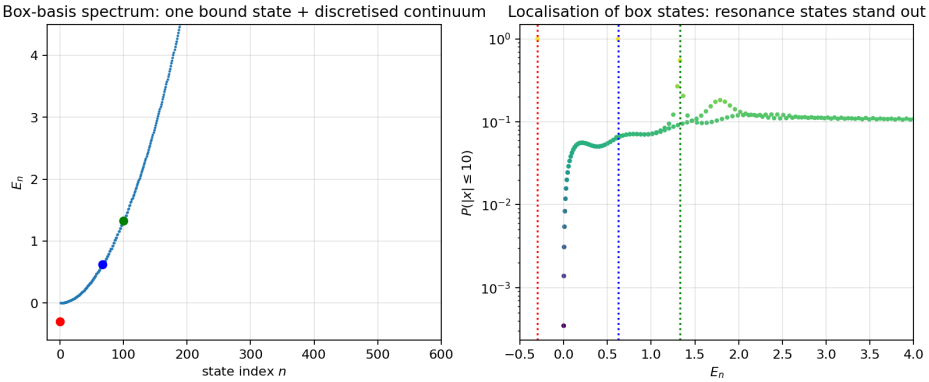
Convergence in J ($L = 150$): $E_0 = 23.49, 5.74, 0.940, -0.1946, -0.29996, -0.30000000$ for $J = 10, 20, 40, 80, 160, 320$ (eigenvalues converge from above – the variational principle). (The Moodle page shows $E_0 = -0.30001954$, presumably from a coarser quadrature; ours is exact to 5×10^{-13}.)

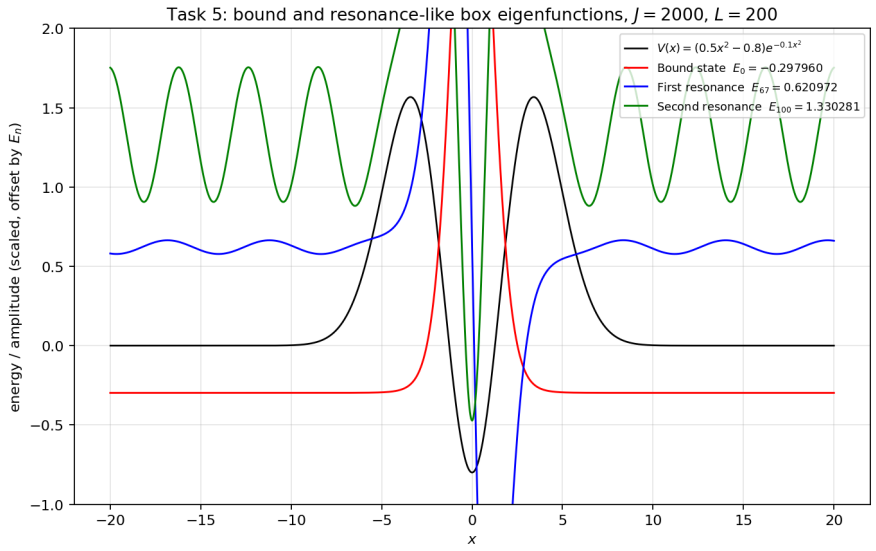
Task 5b — Short-ranged potential $V = (0.5x^2 - 0.8)e^{-0.1x^2}$

$J = 2000$, $L = 200$. One true bound state $E_0 = -0.297960$. The most localised states above threshold are

	n	E_n	$P(\lvert x\rvert \le 10)$
first resonance-like	67	0.620972	0.995
second resonance-like	100	1.330281	0.548

in agreement with the Hermitian peaks 0.620971 and 1.328824; the small differences are the box discretisation error: box-size test (J, L): $(500, 50) \to 0.620814/1.327870$; $(1000, 100) \to 0.620911/1.328930$; $(2000, 200) \to 0.620972/1.330281$; $(3000, 300) \to 0.621036/1.331103$ (the continuum spacing $\sim \pi/L$ pulls the levels around; E_0 is unchanged -0.297960).





10 Lecture 10 — Non-Hermitian Quantum Mechanics and Siegert States

Remark — Why NHQM

NHQM (∼Gamow 1928, Siegert 1939) explains *resonances*. In Hermitian QM resonances appear only as opaque peaks of $|T(E)|^2$: narrow at 0.6209, broader at 1.33 and overlapping ones above. NHQM assigns each of them a complex energy and a wavefunction.

Definition — Siegert states and Siegert boundary conditions

Consider $\big(-\frac{\hbar^2}{2m}\partial_{xx}+V\big)\psi=E\psi$ with $V=0$ outside $[a,b]$ and $E=\hbar^2k^2/2m$.

$$\psi(x\to\pm\infty)\propto e^{\pm \mathrm{i}kx} \iff \big(\partial_x-\mathrm{i}k\big)\psi\big|_{x=b}=0,\quad \big(\partial_x+\mathrm{i}k\big)\psi\big|_{x=a}=0 \quad (*)$$

(purely outgoing waves on both sides). Bound states: $k=\mathrm{i}\kappa$, $\kappa>0$, decaying. NHQM keeps $(*)$ but allows $k=k_R+\mathrm{i}k_I\in\mathbb{C}$, so $E=\frac{\hbar^2}{2m}(k_R^2-k_I^2)+\mathrm{i}\frac{\hbar^2}{m}k_Rk_I$ is complex.

Proof — Real E is impossible for $k>0$ outgoing states

For real E multiply the TISE by ψ^* and its conjugate by ψ and subtract: $\frac{d}{dx}\big(\psi^*\psi'-\psi\psi^{*\prime}\big)=0$, so $F(x)=\psi^*\psi'-\psi\psi^{*\prime}$ is independent of x. If $\psi=Ae^{\mathrm{i}kx}$ $(x>b)$ and $\psi=Be^{-\mathrm{i}kx}$ $(x<a)$, then $F(x>b)=2\mathrm{i}k|A|^2$ and $F(x<a)=-2\mathrm{i}k|B|^2$; equality forces $|A|^2=-|B|^2$, hence $A=B=0$. So Siegert states require **complex** E. ■

Classification. With $k=k_R+\mathrm{i}k_I$: $k_R=0,k_I>0$: **bound states** (L^2); $k_R=0,k_I<0$: anti-bound (virtual) states; $k_R\neq0,k_I<0$: **resonances**, $E=\varepsilon-\frac{\mathrm{i}}{2}\Gamma$, $\Gamma>0$; $k_R\neq0,k_I>0$: anti-resonances (E^* of a resonance). Resonance wavefunctions *explode* $\propto e^{|k_I|x}$ as $x\to\pm\infty$, hence are not in L^2. One can show that bound states, anti-bound states, resonances and anti-resonances (with continuum contributions) form a complete set in a suitable function space; orthogonality needs an appropriate scalar product.

Definition — c-product and Zeldovich regularisation

In Hermitian QM $\langle\psi^{(1)}|\psi^{(2)}\rangle = \int \psi^{(1)*}\psi^{(2)}\,dx > 0$ for $\psi \neq 0$. For a non-Hermitian $\hat{H} = \hat{H}^T \neq \hat{H}^\dagger$ (complex symmetric) one uses the **bilinear (c-)product without complex conjugation**

$$(\psi^{(1)}|\psi^{(2)}) = \int dx\,\psi^{(1)}(x)\psi^{(2)}(x), \quad (\psi|\psi)_{\mathrm{Z}} = \lim_{\varepsilon\to 0^+}\int dx\,\psi^2 e^{-\varepsilon x^2}.$$

It can vanish for $|\psi\rangle \neq 0$ (like the Minkowski metric $\eta_{\mu\nu}x^\mu x^\nu$ for a light-like x), a hallmark of non-Hermiticity. It may be omitted for bound states since those wavefunctions are real.

Warning — Non-Hermitian matrices and exceptional points

For $\underline{H} = \underline{H}^\dagger$ the eigenvalues are real and the eigenvectors orthogonal. For a non-Hermitian matrix the eigenvalues $\mathcal{E} \in \mathbb{C}$ ($\underline{H}\vec{c} = \mathcal{E}\vec{c}$) and **exceptional points** can occur, where eigenvalues *and* eigenvectors coalesce and the matrix is not diagonalisable. Example (complex symmetric): $\underline{H} = \begin{pmatrix} 1 & \mathrm{i} \\ \mathrm{i} & -1 \end{pmatrix}$ has $\det(\underline{H} - \mathcal{E}) = \mathcal{E}^2$, a double eigenvalue $\mathcal{E} = 0$ with the single eigenvector $(1, \mathrm{i})^T$ (and $\underline{H}^2 = 0$). The Hermitian matrix $\begin{pmatrix} 1 & 3+7\mathrm{i} \\ 3-7\mathrm{i} & 2 \end{pmatrix}$ on the other hand is always diagonalisable with real eigenvalues.

11 Lecture 11 — Complex Scaling, Resonances and the Breit–Wigner Profile

11.1 Why we need something computable

Solving $(*)$ directly is a boundary-value problem: pick a trial E (or k), integrate $\psi_{\mathrm{I}} = C_+ e^{\mathrm{i}kx}$ from the right to b, the interior ψ_{II}, match ψ, ψ' at b and a, and demand $\psi_{\mathrm{III}} \propto e^{-\mathrm{i}kx}$; the resulting *transcendental equation for E* can be satisfied only for special E. Since the exploding Siegert functions cannot be handled in a square-integrable basis, we **analytically continue** the wavefunction.

Definition — Complex scaling

Analytic continuation $\psi(x) \mapsto \psi(z)$, $z = xe^{\mathrm{i}\theta}$, $\theta > 0$ (rotation of the coordinate axis by θ):

$$\psi_\theta(x) = \psi(xe^{\mathrm{i}\theta}), \qquad \hat{H}_\theta = -\frac{\hbar^2}{2m}e^{-2\mathrm{i}\theta}\partial_{xx} + V(xe^{\mathrm{i}\theta}), \qquad \hat{H}_\theta\psi_\theta = E\psi_\theta.$$

$\hat{H}_\theta$ is complex symmetric, $\underline{H} = \underline{H}^T$, $\underline{H} \neq \underline{H}^\dagger$, and $V(xe^{\mathrm{i}\theta})$ must be analytic (true for our Gaussian-type V). The spectrum is *independent of* θ for the discrete (bound-state and resonance) part.

Theorem — When does the resonance become square integrable?

Let E be a resonance with $k = k_R + \mathrm{i}k_I$, $k_R > 0$, $k_I < 0$. Then $e^{\mathrm{i}kxe^{\mathrm{i}\theta}} \sim e^{F_\theta x}$ with $F_\theta = |k_I|\cos\theta - k_R\sin\theta$. At $\theta = 0$, $F_0 = |k_I| > 0$ (explodes); $F_\theta < 0$ (decays, $\psi_\theta \in L^2$) iff

$$\theta > \theta_{\mathrm{crit}}, \qquad \tan\theta_{\mathrm{crit}} = \frac{|k_I|}{k_R}, \qquad \text{equivalently } 2\theta_{\mathrm{crit}} = |\arg E_{\mathrm{res}}|.$$

Meanwhile the real continuum $E = \hbar^2k^2/2m$ is rotated onto the rays $\arg E = -2\theta$, so $\hat{H}_\theta$ in a square-integrable basis has: real bound states (unchanged), **continuum rays** rotating

by -2θ, and stationary complex resonance eigenvalues $E_r = \varepsilon - \frac{\mathrm{i}}{2}\Gamma$ that are uncovered once $\theta > \theta_{\text{crit}}$.

Method — Complex-scaling calculation (Project part 6)

1. Build the box-basis matrix $H^{(\theta)}_{jj'} = e^{-2\mathrm{i}\theta} T_{jj'} + \int \varphi_j V(xe^{\mathrm{i}\theta})\varphi_{j'} dx$ (same basis as Task 5).
2. Diagonalise (general complex eigensolver, not `eigh`) for $\theta = 0, 0.05, 0.10, 0.15, 0.20, 0.25$ $(, 0.30)$.
3. Plot all eigenvalues in the complex E plane; identify the points that *do not move* with θ.
4. Read off $E_r = \varepsilon - \frac{\mathrm{i}}{2}\Gamma$ and compare the Lorentzian (**Breit–Wigner**) profile $|T(E)|^2 \simeq \dfrac{(\Gamma/2)^2}{(E-\varepsilon)^2 + (\Gamma/2)^2}$ with the Hermitian $|T(E)|^2$ of Task 3.
5. Continue the eigenvector back: $\psi(x) = \psi_\theta(xe^{-\mathrm{i}\theta})$; on the real axis this is the outgoing solution that grows like $e^{|k_I|x}$.

Proposition — Physical meaning of a resonance

If the initial packet lies in the interaction region, $\Psi(t,x) \approx \psi_{\text{res}}(x)\, e^{-\mathrm{i}(\varepsilon - \mathrm{i}\Gamma/2)t/\hbar}$, so

$$|\Psi(t,x)|^2 = |\psi_{\text{res}}(x)|^2 e^{-\Gamma t/\hbar} : \qquad \text{exponential decay with lifetime } \tau = \hbar/\Gamma.$$

This explains why the narrow 0.6209 resonance ($\Gamma \sim 10^{-4}$) traps the wavepacket for a very long time.

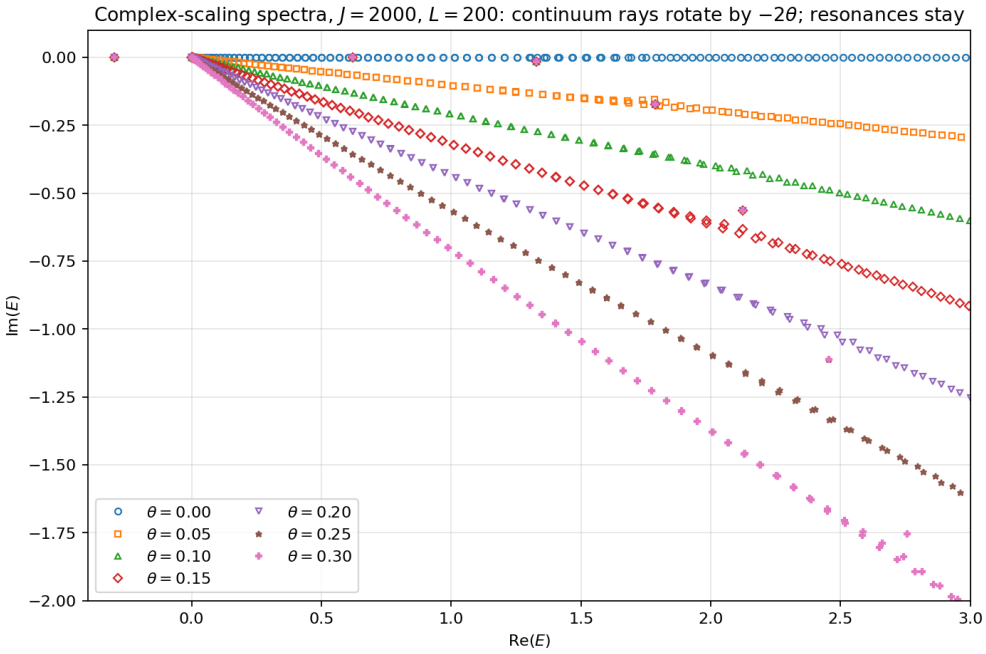
Remark — Analytic continuation (recap)

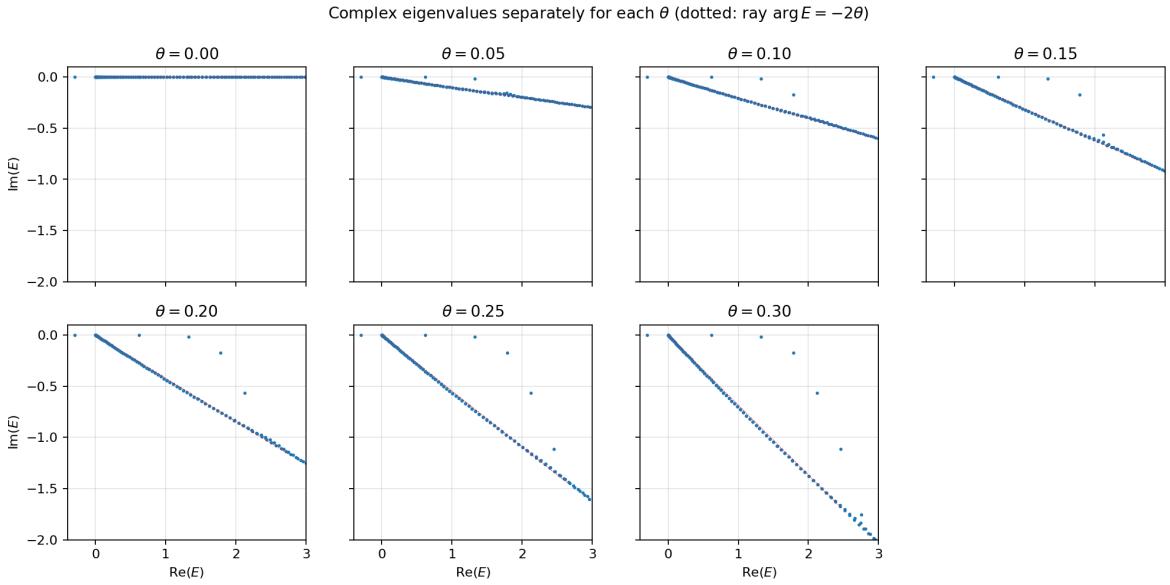
$f(x) = x^3$ or $\sin x$ extends to $f(z) = z^3$, $\sin z = (e^{\mathrm{i}z} - e^{-\mathrm{i}z})/2\mathrm{i}$, $e^{\mathrm{i}z} = \sum z^n/n!$ for $z = x + \mathrm{i}y$, by replacing $x - x_0$ with $z - z_0$ in the Taylor series $f(x) = \sum_n f^{(n)}(x_0)(x - x_0)^n/n!$. The wavefunction is continued the same way, so $\psi(xe^{\mathrm{i}\theta})$ makes sense for $\theta > 0$.

11.2 Project Task 6 — Complex Scaling (NHQM)

Method. $\hat{H}_\theta = -\frac{1}{2}e^{-2\mathrm{i}\theta}\partial_{xx} + V(xe^{\mathrm{i}\theta})$ in the same box basis ($J = 2000$, $L = 200$), complex eigensolver, $\theta = 0, 0.05, \ldots, 0.30$.

Task 6a — Spectra





The continuum points fall on rays $\arg E = -2\theta$; the bound state ($E_0 = -0.297960$) and five isolated resonances stay fixed.

Task 6b — Resonance energies (θ-stable eigenvalues)

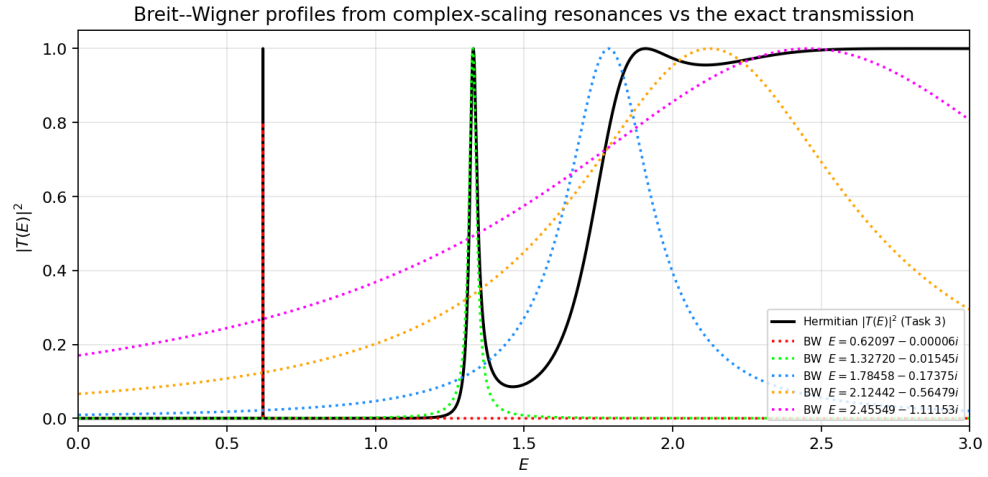
ε	$-\Gamma/2$	Γ	$1/\Gamma$
0.620971	-0.0000583	1.165×10^{-4}	8.58×10^{3}
1.327197	-0.015447	3.089×10^{-2}	32.4
1.784583	-0.173751	0.3475	2.88
2.124422	-0.564795	1.130	0.885
2.455486	-1.111532	2.223	0.450

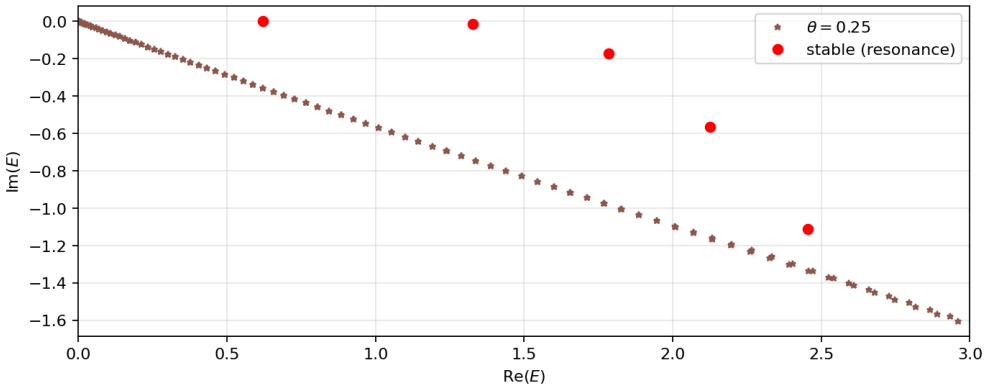
The three lowest agree between $\theta = 0.20, 0.25, 0.30$ to 10^{-12}; and the five values coincide

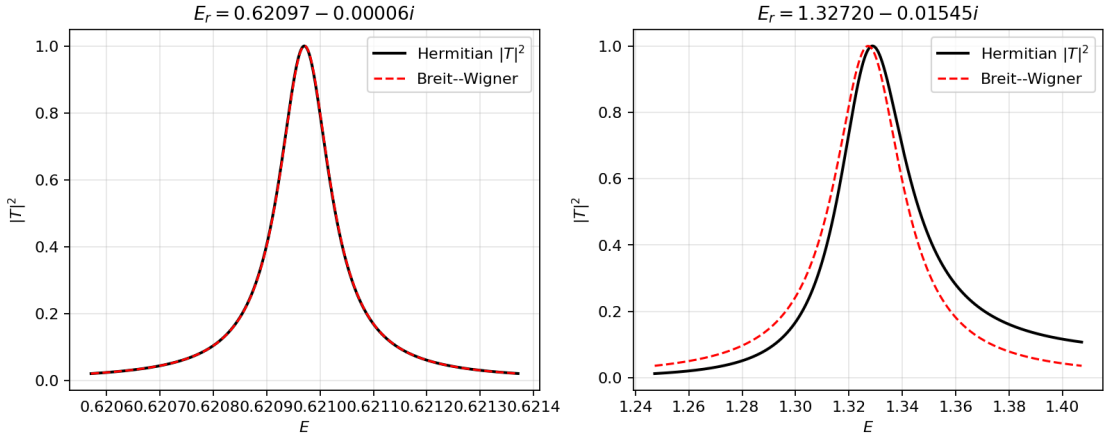
with those shown on the Moodle Breit–Wigner plot. Consistency with Task 3: FWHM of the Hermitian peaks 1.165×10^{-4} and 3.20×10^{-2} vs $\Gamma = 1.165 \times 10^{-4}$ and 3.09×10^{-2}; the small offset for the second peak reflects that a Breit–Wigner form is only an approximation (background and asymmetry).

Task 6c — Breit–Wigner profiles

$|T(E)|^2 \simeq (\Gamma/2)^2/[(E-\varepsilon)^2 + (\Gamma/2)^2]$ for each resonance versus the Hermitian $|T|^2$ of Task 3:



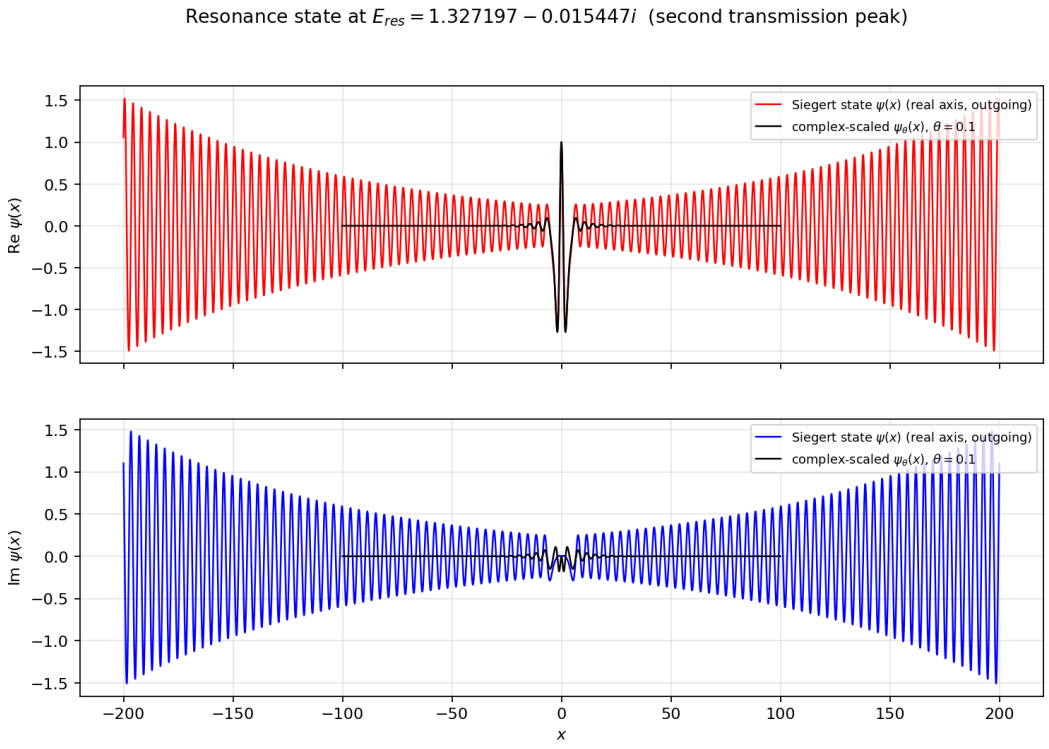




The two narrow resonances are reproduced almost perfectly; the broad ones (overlapping, complex Γ comparable to ε) show only qualitative Lorentzian envelopes: interference between overlapping resonances and the background breaks the isolated-resonance picture.

Task 6d — Resonance wavefunction of the second peak

$E_{\rm res} = 1.327197 - 0.015447\,\mathrm{i}$; the complex-scaled eigenfunction at $\theta = 0.1$ is even and square integrable (edge amplitude 4×10^{-15} of the maximum). The outgoing (Siegert) solution obtained on the real axis by finite-difference integration outward from $x = 0$ with the same complex energy grows like $e^{|k_I|x}$ with $k = 1.629259 - 0.009481\,\mathrm{i}$; numerically $|\psi(200)|/|\psi(50)| = 4.146 = e^{0.009481 \cdot 150}$. The two agree near the origin (relative deviation 2.5×10^{-4} for $|x| < 8$ after continuing along the ray $e^{\mathrm{i}\theta}$).



12 Consolidated Summary and Self-Check

Key Formula Summary — The whole course on one page

Free evolution	$\Psi(t,x) = \frac{1}{\sqrt{2\pi\hbar}} \int dp\, \phi(p) e^{\mathrm{i}px/\hbar} e^{-\mathrm{i}p^2 t/2m\hbar}$
Delta function	$\frac{1}{2\pi} \int dx\, e^{\mathrm{i}\xi x} = \delta(\xi)$, $\delta_L = \sin(L\xi)/\pi\xi$
Scattering BC	$\psi_p(+\infty) = T\psi_p^{\rm F}$, $\psi_p(-\infty) = \psi_p^{\rm F} + R\psi_{-p}^{\rm F}$, $\lvert T\rvert^2 + \lvert R\rvert^2 = 1$
Completeness	$\sum_j \varphi_j \varphi_j^* + \int dp\, \psi_p \psi_p^* = \delta(x - x')$
Møller/$\hat{S}$	$\hat{S} = \lim e^{\mathrm{i}\hat{H}_0 T} e^{-2\mathrm{i}\hat{H}T} e^{\mathrm{i}\hat{H}_0 T}$
LSE	$\lvert\psi_p^{\pm}\rangle = \lvert\psi_p^{\rm F}\rangle + (E_p - \hat{H}_0 \pm \mathrm{i}\varepsilon)^{-1} \hat{V} \lvert\psi_p^{\pm}\rangle$
S-matrix	$\langle\psi_p^{\rm F}\rvert\hat{S}\lvert\psi_{p'}^{\rm F}\rangle = \delta(p - p') - 2\pi\mathrm{i}\delta(E_p - E_{p'}) \langle\psi_p^{\rm F}\rvert\hat{T}\lvert\psi_{p'}^{\rm F}\rangle$
T, R	$T = 1 - 2\pi\mathrm{i}\frac{m}{\lvert p\rvert}\langle\psi_p^{\rm F}\rvert\hat{V}\lvert\psi_p^{+}\rangle$, $R = -2\pi\mathrm{i}\frac{m}{\lvert p\rvert}\langle\psi_{-p}^{\rm F}\rvert\hat{V}\lvert\psi_p^{+}\rangle$
Resonance	$E = \varepsilon - \frac{\mathrm{i}}{2}\Gamma$, $\lvert T\rvert^2 \simeq \frac{(\Gamma/2)^2}{(E-\varepsilon)^2 + (\Gamma/2)^2}$, $\tau = \hbar/\Gamma$
Complex scaling	$\hat{H}_\theta = -\frac{\hbar^2}{2m} e^{-2\mathrm{i}\theta} \partial_{xx} + V(xe^{\mathrm{i}\theta})$, $\tan\theta_c = \lvert k_I\rvert/k_R$

Remark — Organisation choices and clarifications made in this note

- The lectures use $\hbar$ explicitly on the boards and $\hbar = m = 1$ in the project; both are kept and switched only where stated.
- The board writes $\mathcal{T}(p)$ and $\mathcal{R}(p)$ in L9 and $T(p), R(p)$ in L4/L7; they are the same (shown by a direct comparison in Section 9).
- The iε sign: $+\mathrm{i}\varepsilon$ yields ψ^+ (in-state, retarded), $-\mathrm{i}\varepsilon$ yields ψ^- (out-state, advanced).
- Task 2's test function $\xi e^{-10\xi^2}$ is odd, so its integral against the even δ_L is trivially zero; an even-part test function was added to display convergence.
- Contents from the advanced reference (Siegert pseudostates, general Wigner delay, etc.) that are not in L1–L11 are intentionally omitted.

Warning — Common pitfalls

- Do not normalise scattering states ψ_p to 1: they are δ-normalised, $\langle\psi_p|\psi_{p'}\rangle = \delta(p - p')$.
- $e^{-\mathrm{i}\hat{H}t}e^{\mathrm{i}\hat{H}_0 t} \neq e^{-\mathrm{i}(\hat{H}-\hat{H}_0)t}$ for non-commuting operators.
- In the box basis the many box levels above $E = 0$ are *not* resonances; only strongly localised ones are.
- For complex scaling use `eig`, not `eigh`; the matrix is complex *symmetric*, not Hermitian.

Warning — Limitations of this study

- Task statements 3–6 were inferred from lectures and screenshots (see the remark at the start); parameters (Δx, σ_p, x_0, θ list) follow the Moodle pictures where visible, otherwise are my choices. Moodle's Task 4 uses $\sigma_p = 10^{-6}$ on a $\pm 4 \times 10^6$ domain; here $\sigma_p = 2 \times 10^{-5}$ with the scattering-state method was used instead.
- The Task 3 lattice result is exact only up to $O(\Delta x^2)$; near the 10^{-4}-wide resonance this is a relative shift $\sim 10^{-2}$ of the peak width.
- Box discretisation (Task 5) shifts the resonance-like levels by $\lesssim 10^{-3}$; use complex scaling (Task 6) for converged ε, Γ.